

Nonlinear Device Measurement, Characterization, and Modeling for High Power RF Applications

DISSERTATION

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ABSTRACT

To satisfy power and frequency requirements for commercial RF applications, various semiconductor materials and device structures have been introduced and developed rapidly. These semiconductors need to handle modulated signals exhibiting high peak to average power ratio (PAPR) and wide modulation bandwidth to meet the requirements of modern wireless standards. In this context, measurements with continuous wave (CW) single tone or two-tone excitations are no longer sufficient to characterize power amplifiers (PAs) for such applications due to the inherent memory effects present in PAs. In this thesis a new testbed is developed with a Large Signal Network Analyzer (LSNA) to perform multi-harmonic broadband measurements for periodically modulated signals with bandwidth exceeding the IF receiver bandwidth of the LSNA.

Also a new pulsed real-time active load-pull (RTALP) measurement system which is developed and used to characterize devices under iso-memory effect is reported in this thesis. With the LSNA system, the multiple recording technique are combined in order to reduce the long measurement time and bypass the pulse desensitization problem. With this novel combined technique, output power and gain at 1 dB compression gain plane are extracted from a double power-sweep measurements of the input and output power levels.

It is necessary for successful device modeling to account for memory effects such as self-heating, traps in HEMTs and parasitic bipolar junction transistor in FETs. Due to the slow time response of these memory effects, fast pulsed-IV/pulsed- RF measurements have

been used to maintain constant device temperature, body BJT voltage in FETs or trap state in HEMTs. Therefore, new microwave characterization techniques are also presented to characterize these unwanted memory effects by estimating the internal device operating temperature and extracting the trap relaxation time-constants in AlGaIn/GaN HEMTs.

Having characterized results under the developed measurement system, a new measurement-based model with artificial neuron network (ANN) is finally proposed and applied to a SOI MOSFET device. The verification results demonstrate that the time-consuming measurement process can be dramatically reduced by using RTALP measurement data, and that fairly accurate large-signal RF device model can be easily extracted from these measurements using the ANN approach.

This is dedicated to my family and friends whom I love.

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CHAPTER 1

INTRODUCTION

1.1 Measurement System Development for Nonlinear Device Characterization

1.1.1 Multi-Harmonic Broadband Measurement System Using a Large Signal Network Analyzer

The growing demand for wireless communication services is motivating the development of power amplifiers (PAs) operating with wider bandwidth. Therefore these amplifiers must also handle modulated signals with large peak to average power ratio (PAPR) as specified by modern wireless standards. Measurements with continuous wave (CW) single-tone and two-tone excitations are not sufficient to characterize and model PAs for such applications due to the inherent memory effects in PAs. Nonlinear or large-signal network analyzers (NVNA/LSNA) offer promising possibilities for characterizing the nonlinear behavior of PAs for CW and also periodically-modulated excitations. Using an analog sub-sampling technique, the LSNA can acquire multitone periodic signals with wide RF bandwidth (up to 50 GHz in the LSNA) and effectively remapped them under appropriate conditions within the IF bandwidth of the receiver (IF bandwidth is 10 MHz in Maury's LSNA and 100 MHz in VTS's SWAPP). Once downconverted the RF tones are then digitized using analog-to-digital converters (ADCs) after the low-pass filter. In commercialized

LSNAs, the modulation bandwidth for the RF signals acquired is limited to twice the IF modulation bandwidth, e.g. 20 MHz in the LSNA. It is however possible under appropriate conditions to increase further the modulation bandwidth, by choosing the measurements parameters of the LSNA such as the ADC frequency f_{ADC} , LO frequency f_{LO} , and the number of FFT in such a way that none of the fundamental and harmonic tones fold down to the same IF frequency during down-conversion [1] [2] [3].

Such a broadband measurement system was realized within a large-signal network analyzer by adding an additional frequency synthesizer to set the ADC clock and a counter driven by the ADC clock for triggering the measurements (see Figure 2.1). Once calibrated, the modulated RF signals are acquired in a single LSNA measurement. No stitching of measured data from multiple measurements is therefore required [4] [5], to obtain the amplitude and phase of the various tones of the multi-harmonic wide-band periodically modulated signal acquired. The triggering frequency is set to a sub-harmonic of the greatest common divisor frequency f_{gcd} of all the down-converted IF tones. This frequency is equivalent to the IF period used in the four tone calibrated broadband measurement recently reported [6]. This trigger guaranties that the phase relationship between each of tones remains the same in successive measurements. A phase repeatability of a fraction of a degree is achieved in successive calibration measurements.

Chapter 2 presents the calibration technique and the measurement results including spectral regrowth components for a 480W GaN Doherty power amplifier excited by a multisine with 6 dB PAPR. A conventional LSNA calibration including SOLT, power meter, and harmonic phase reference is used for the RF fundamental and harmonics. The IF phase

calibration is then performed at each RF harmonic band of interest using a reference modulated RF source. A spectrum analyzer can also be used to compensate for the power difference between the targeted and measured power in the multitone signal. The measurement results reported here are limited to about 80 MHz bandwidth, 63 modulation tones, and fundamental and second harmonic RF frequencies. However the system can be extended to wider bandwidth, more modulation tones and larger number of RF harmonics as needed, provided a reference modulated RF source is available for the modulation calibration.

1.1.2 Pulsed-RF Real-Time Active Load-Pull Technique with a Passivated AlGaIn/GaN HEMT Using a Large Signal Network Analyzer

As the complexity of commercial signals such as LTE is growing to address the increasing need for bandwidth and user capacity in wireless communication, the peak to average power ratio (PAPR) is also increasing reaching 10 dB for LTE. As the PAPR is increased, the operation point of power amplifiers (PA) must be proportionally backed off from their own 1 dB compression gain point in order to avoid hard non-linearities. Thus at peak power the PAs will be operating at their 1dB compression while keeping an acceptable signal distortion.

It results that the 1 dB compression gain point remains of great importance in the evaluation of the transistor linearity and the design of power amplifiers [7]. To determine the 1dB compression for a fixed load impedance a power sweep measurement (multiple measurements) is required. Considering that at least hundreds to thousands different impedance points should be measured to generate the P1dB load impedance contour plot, this is usually a very measurement intensive process. A similar large number of measurements is also required if the maximum power efficiency impedance termination is targeted for different

output powers. Clearly the measurements required to determine the optimal load conditions for maximum output power under linear or power-efficient operation constraints are quite time consuming and this greatly impact the PA design cycle. A time efficient scheme would therefore be greatly beneficial considering that ideally P1dB load impedance contour plot measurements should be available for multiple biasing conditions or multiple modes of operation (harmonic termination).

Passive load-pull using mechanical tuners is the most common technique used to find the optimal load impedance contours yielding the maximum output power and efficiency in transistors. However, passive load pull systems suffer from several limitations. First of all, measurement with passive load pull systems is typically a very time-consuming process due to the use of mechanical tuners. Second, passive tuners are limited in term of the maximum reflection coefficients they can attain due to the tuners' inherent loss [8]. To address this issue, active load pull (ALP) systems based on the closed-loop architecture [9], [10] have been developed to improve the reflection coefficient attainable. However, due to potential stability problem, a complex controlling scheme is required and closed-loop ALP systems are not extensively used. The stability issue was eliminated with the development of the open-loop ALP architecture [11]. However the long measurement time issue still remains an issue. This has been resolved with With the advent of the real-time active loadpull (RTALP) measurement technique [12] it is possible to address all the issues. RTALP can dramatically reduce the required measurement time and there is no limitation in the attainable reflection coefficient without the risk of stability issue. This approach is thus considered to be highly promising to facilitate the design of linear power-efficient PAs when used in combination with nonlinear vector network analyzer (NVNA).

For successful device modeling and characterization it is also essential to account for slow memory effects such as self-heating, traps in HEMTs and parasitic bipolar transistor in SOI or SOS MOSFETs which are originating from various physical processes in the transistors [13]. Due to the slow time response of these memory effects, fast pulsed-IV/pulsed-RF measurements have been used to maintain constant device temperature, body BJT voltage in FETs or trap state in HEMTs [14]. Several studies using pulsed RF measurements have been reported. A pulsed VNA was developed for a pulsed ALP measurement configuration to detect the pulsed-RF power within a given pulse width [15]. A pulsed loadpull system using a digital scope and a power meter was also introduced to measure the power gain of devices under pulsed condition [16]. In addition, a pulsed ALP system with an LSNA was used to design high-efficiency Class-B RF PAs with AlGaIn/GaN HEMTs in [17]. The previous studies in the pulse measurement system, however, were subject to the boundary of ALP configuration, which still require long measurement times in order to search for an appropriate load condition for the targeted applications. Moreover, it is well known that pulse measurements with VNAs suffer from a significant loss of dynamic range as the duty cycle of the pulse decreases. This is characterized by the desensitization factor:

$$\text{Desensitization} = \frac{P_{peak}}{P_{fund}} = 20 \times \log(\text{Duty_rate}). \quad (1.1)$$

where P_{peak} is the peak instantaneous RF power of the signal measured and P_{fund} the average RF power at the fundamental measured by the test equipment. To compensate for the loss of dynamic range, a modified architecture was introduced with the LSNA, in which all of the tones within the main lobe of the pulse signal are acquired to keep 90 % of the power [14]. The desensitization is then reduced to

$$\text{Desensitization} = 10 \times \log(\text{Duty_rate}). \quad (1.2)$$

Nevertheless, this approach requires some signal processing and is still restricted to duty cycle above 0.3 %. Indeed the total RF power is spread into multiple tones of amplitude proportional to the desensitization and eventually the power of each tone will be fall below the minimum detectable level of the system as the duty cycle decreases. Recently a new approach, the multiple recording technique, was demonstrated for the LSNA which enables one to measure a pulsed signal for arbitrarily low duty cycles without any dynamic range reduction [18], [19]. There is thus no desensitization associated with multiple recording:

$$\text{Desensitization} = 0 \text{ dB.} \quad (1.3)$$

Indeed in this technique, a pulse control board inside the LSNA control the analog digital converter(ADC) acquisition so that the sampled down-converted RF data are only acquired when the peak RF pulse is applied to the DUT. It can be experimentally verified that the data acquisition in the modified LSNA does not exhibit any degradation in dynamic range as the pulse duty cycle decreases [20]. The limitation in the pulse-width is itself only limited itself by the latency of the low-pass filter used after the sampler in the LSNA.

In Chapter 3, the RTALP and the multiple recording techniques will be combined in order to reduce the exhaustive measurement time associated with the loadpull as well as to solve the desensitization problems associated with the pulse excitation. With this combined technique, the P1dB compression gain extraction was conducted to find the optimal load condition for the output power and gain on the entire Smith Chart. For the sake of verification of the new technique proposed in this dissertation, a passivated AlGaIn/GaN HEMT was selected as a device-under-test (DUT). Silicon nitride(SiN) passivation and field plates will be used in this device to minimize the unwanted memory effects on the HEMT devices. The selected DUT is therefore expected to yield fairly similar measurement results

in both continuous wave (CW) and pulsed excitations. Furthermore, the measurement results obtained under the two different measurement schemes (RTALP, Constant Phase) will be compared for three critical load terminations: maximum output power, maximum gain, and matched load (50 ohms).

1.2 Memory Effects Characterization

1.2.1 New Thermometry and Trap Relaxation Characterization for AlGaIn/GaN HEMTs Using Pulsed-RF Excitations

AlGaIn/GaN HEMTs have demonstrated impressive performance for high power RF amplification [21] [22]. However they can exhibit temporary current collapse or IV knee walkout which degrade their RF performance [23], [24] [25]. Many studies have been reported which ascribe such performance reductions to various memory effects such as self-heating, surface states and bulk traps [22]. Although many solutions have been devised to mitigate these memory effects, important RF dispersion effects can still be observed in AlGaIn/GaN HEMTs particularly under high RF voltage operation. The proper understanding and characterization of these memory effects is thus of critical importance for modern digital communication which feature modulated RF signals with high peak-to-average power-ratio (PAPR). New large-signal pulsed-RF measurement-techniques are presented in chapter 4 to address this issue.

1.2.2 New Trends for the Nonlinear Measurement of High Power RF Transistors and Amplifiers with Memory Effects

Various type of low-frequency memory effects arise in transistors due to different physical processes. Self-heating, trapping, and bipolar parasitic transistor effects in MOSFET

are typically the main sources of low-frequency memory effects with electrical time constants running from 100 s to 100 ns. It is of interest to investigate the impact of these memory effects on the performance of PAs excited by signals with high PAPR. This investigation of the dynamics of the memory processes taking place will permit us insights in the practical requirements for accurately characterizing and modeling RF transistors.

Complex modulated signals such as WiMAX and LTE exhibit large PAPR reaching 10 dB. However PAPRs larger or equal to 10 dB arise with a very small probability (typically 10^{-5}) and take place for short duration on the order of 100 ns. Thus most of the time the amplifier is amplifying the modulated signals at the average power. Yet the amplifier still needs to amplify quasi-linearly the envelope peaks with a PAPR up to 7 dB to maintain a reasonable bit error rate. Furthermore the amplifiers must be designed to operate with a good efficiency from the average power to the peak power to deliver acceptable average power efficiency. To successfully complete this complex amplifier design, the PA designers must definitively rely on simulation and measurement data which provide a realistic evaluation of the device performance.

Load-pull is an experimental technique which is often used either in simulation or experimentally to design power amplifiers. In load-pull measurements the multi-harmonic terminations presented by the load and source networks are tuned to optimize the power added efficiency (PAE) of the PA while maintaining its gain over the dynamic range of the output power. Conventionally load-pull relies on constant wave (CW) RF signals. However CW load-pull may not provide a reliable evaluation of the transistor performance if the device exhibits strong memory effects. To demonstrate this let us first consider self-heating. For amplifiers operating in class B, AB or class F, the power dissipated by the transistor increases with increasing input power. However considering that peak power is

only achieved for a very short duration (100 ns) and has a low probability of occurrence and thermal effects are usually relatively slow ($\sim \mu\text{s}$), the device will remain operating at the temperature established by the average dissipated power. Therefore CW load-pull at the peak power may not yield realistic results due to the comparatively much larger self-heating it will induce. Next consider the impact of traps in devices such as GaN HEMTs. Under low-power CW operation the drain voltage peak remain typically small and the traps are not activated. However for modulated RF signals, the instantaneous drain voltage reaches intermittently large values (50 to 60 V). Traps in GaN typically feature very short capture time (ns) at high drain voltages and very long emission time (s) at low drain voltages. It results that traps gradually charge during the brief the periods of peak power and do not have time to discharge during the intervening time. The dominant response of the transistor for the average input power is then directly affected by the charged traps and associated knee walkout effect resulting from these intermittent high power peaks of the modulated RF signal. Thus the dynamics of charge trapping for modulated RF excitations cannot be captured by CW load-pull simulations and measurements.

An increase of output 1/f noise can also be detected due the activated traps. Note that the state of the traps is not determined by the DC bias condition but by the RF load line (cyclostationary effect) [26] [27] [28]. Other memory effects such as the slow parasitic bipolar in SOI-MOSFET may also be expected to be affected by the dynamic RF load lines.

1.3 Nonlinear Device Modeling Based on Artificial Neural Network (ANN)

Various semiconductor materials (Si, GaAs, GaN, InP, etc) and device structures (SOI-MOSFET, MESFET, LDMOS, HEMT, etc) have been introduced and rapidly deployed in order to meet the high power and high-efficiency requirements of modern high frequency wireless communication [26] [29]. Under such circumstances, conventional modeling approaches (e.g., physically based or analytical closed-form empirical models) cannot keep up with the fast pace of technology development [30]. Measurement-based models, which by-pass this technology dependence in the model equations, were first introduced with linear measured S-parameters data and later on with nonlinear device models like the Root model and with nonlinear behavioral models such as X-parameters [31]. In all these model the functional dependence is typically implemented using spline functions. These measurement-based modeling approaches facilitate the development of large signal models and thus accelerate the design cycle for power amplifiers [32] [33] [34]. However, the main limitation in the use of spline function is (1) their poor extrapolation outside of the range of the data available and (2) their possible oscillation between data-points. The recently introduced artificial neural network (ANN) method provides very smooth functions and non-vanishing derivatives of infinite order while guaranteeing a capability to fit any non-linearities of devices [35] [36] [37].

However, the model development cycle is still restricted due to the large number of S-parameter or nonlinear measurements needed for the model extraction. Indeed the measurement time rapidly increases as a dense grid on the intrinsic voltage plane is required for a better extraction of the model. The advent and successful deployment of non-linear vector measurement systems has induced an interest in using large signal characterization instead

of conventional small signal characterization for device modeling. The main advantage of using large signal measurements for device modeling is that the model parameters are extracted under real device operating conditions and that the number of needed measurements can be reduced dramatically. The first trial using large signal measurements was conducted for the extraction of the nonlinear charges of a FET [38]. In [34] and [38], the gate and drain currents and charges extracted from large signal measurements were compared to the ones obtained from small signal measurements using an ideal analytical model to show the validity of the large signal approach. In addition to the excellent agreement obtained with the small signal results, the needed number of measurements was found to be reduced from N^2 to N times while obtaining the same intrinsic voltage coverage. However, the time consuming tuner characterization and implementation required in passive load-pull or the large number of amplitude and phase sweep measurements in the active load-pull system remain still a key sticking point in the model development cycle.

In this paper we present a measurement-based quasi-static non-linear model for an SOS-MOSFET whose intrinsic nonlinear charges and current sources are extracted and verified from large signal measurements performed using the real-time active load-pull technique (RTALP) [39] [12]. Because RTALP data typically covers a wide range of device operating point, the model can be extracted from only a single large signal measurement with an LSNA.

By directly comparing the measured and modeled load-lines, the proposed methodology allows for both extremely fast model extraction and verification procedures. Also, by comparing the output power and PAE in the whole Smith Chart, the model ability to guarantees the excellent interpolation as well as extrapolation capabilities.

In addition the FET model is extended to include the parasitic-bipolar junction transistor (P-BJT) of the SOS-MOSFET so as to account for the associated low frequency memory effects and yield a model with improved accuracy.

1.4 Research outline and new contributions

The research presented in this dissertation encompass the following targets. First, a new multi-harmonic broadband measurement system and a new pulsed-RTALP measurement systems will be developed to investigate the benefit of iso-memory and broadband large-signal RF measurements for the characterization of nonlinear devices. Then the characterization of memory effects such as thermal and trap effects in HEMTs and parasitics bipolar transistors in FETs will be pursued for RF power amplification applications. Finally, a measurement based model for the large-signal circuit simulation will be developed from the iso-memory large-signal measurements acquired using an ANN. The ANN model will then be verified for its ability to predict new measurement results.

In Chapter 2, the new contributions are as follows:

1. A multi-harmonic broadband measurement system is developed with an LSNA for modulated signals with bandwidth exceeding the IF receiver bandwidth of the LSNA.
2. An additional IF phase calibration technique is developed and used for the first time for each RF harmonic band of interest using a reference modulated RF source.
3. Even though the measurement results demonstrated are limited to about 80 MHz bandwidth, the system can be used for wider bandwidth, more modulation tones and larger number of RF harmonics as needed.

In Chapter 3, the new contributions are as follows:

1. A new pulsed real-time active load-pull (RTALP) measurement system is developed and characterized using an AlGaIn/GaN HEMT.
2. The multiple recording technique and RTALP system are combined together in order to reduce the long load pull measurement time and to bypass the pulse desensitization problem.
3. The output power and power gain at the 1 dB compression gain plane are extracted from the power sweep measurements using different input power.

In Chapter 4, the new contributions are as follows:

1. A new novel microwave characterization technique is developed to estimate the internal device operating temperature and to measure the trap relaxation time-constants in AlGaIn/GaN HEMTs.
2. Using illumination is further verified to accelerate the emission process confirming the trapping activity and identifying the trap energy levels.
3. For the first time, it is verified that the higher RF load impedances for the same biasing, strongly increase the trapping activity as the instantaneous drain voltage reaches higher peak values.

In Chapter 5, the new contribution is as follow:

1. For the first time, a realistic representation of the peak power response of transistors excited with modulated signals with high PAPR is extracted under pulsed-RF load-pull from the proper hot bias condition.

In Chapter 6, the new contributions are as follows:

1. For the first time, a measurement-based ANN model is fully extracted from large-signal characterization based on measurements acquired using the RTALP technique.

2. Using the various load-line measured for both model extraction and model verification, the proposed methodology enables for an extremely fast extraction procedure without a significant reduction in the model accuracy.

CHAPTER 2

MULTI-HARMONIC BROADBAND MEASUREMENT USING A LARGE SIGNAL NETWORK ANALYZER

The growing demand for wireless communication services is motivating the development of power amplifiers (PAs) operating with wider bandwidth. Therefore these amplifiers must also handle modulated signals with large peak to average power ratio (PAPR) as specified by modern wireless standards. Measurements with continuous wave (CW) single-tone and two-tone excitations are not sufficient to characterize and model PAs for such applications due to the inherent memory effects in PAs. Nonlinear or large-signal network analyzers (NVNA/LSNA) offer promising possibilities for characterizing the nonlinear behavior of PAs for CW and also periodically-modulated excitations. Using an analog sub-sampling technique, the LSNA can acquire multitone periodic signals with wide RF bandwidth (up to 50 GHz in the LSNA) and effectively remapped them under appropriate conditions within the IF bandwidth of the receiver (IF bandwidth is 10 MHz in Maury's LSNA and 100 MHz in VTS's SWAPP). Once downconverted the RF tones are then digitized using analog-to-digital converters (ADCs) after low-pass filtering. In commercialized LSNAs, the modulation bandwidth for the RF signals acquired is limited to twice the IF modulation bandwidth, e.g. 20 MHz in the LSNA. It is however possible under appropriate conditions to increase further the modulation bandwidth, by choosing the measurements

parameters of the LSNA such as the ADC frequency f_{ADC} , LO frequency f_{LO} , and the number of FFT in such a way that none of the fundamental and harmonic tones fold down to the same IF frequency during down-conversion [2].

Such a broadband measurement system was realized within a large-signal network analyzer by adding an additional frequency synthesizer to set the ADC clock and a counter driven by the ADC clock for triggering the measurements (see Figure 2.1). Once calibrated, the modulated RF signals are acquired in a single LSNA measurement. No stitching of measured data from multiple measurements is therefore required [4] [5], to obtain the amplitude and phase of the various tones of the multi-harmonic wide-band periodically modulated signal acquired. The triggering frequency is set to a sub-harmonic of the greatest common divisor frequency f_{gcd} of all the down-converted IF tones. This frequency is equivalent to the IF period used in the four tone calibrated broadband measurement recently reported [6]. This trigger guaranties that the phase relationship between each of tones remains the same in successive measurements. A phase repeatability of a fraction of a degree is achieved in successive calibration measurements.

This chapter presents the calibration technique and the measurement results including spectral regrowth components for a 480W GaN Doherty power amplifier excited by a multisine with 6 dB PAPR. A conventional LSNA calibration including SOLT, power meter, and harmonic phase reference is used for the RF fundamental and harmonics. The IF phase calibration is then performed at each RF harmonic band of interest using a reference modulated RF source. The measurement results reported here are limited to about 80 MHz bandwidth, 63 modulation tones, and fundamental and second harmonic RF frequencies. However the system can be extended to wider bandwidth, more modulation tones and larger

number of RF harmonics as needed, provided a reference modulated RF source is available for the modulation calibration.

2.1 Principle of Phase Calibration

Conventional LSNA bandwidth is limited to 20 MHz. It can be extended to handle an arbitrarily wide modulation bandwidth with a unique calibration algorithm. Using sub-sampling down-conversion [3], an injected modulated RF signal can be downconverted into IF bandwidth by mixing with multiple of harmonics of f_{LO} :

$$\begin{aligned}
& 4 \sum_q \cos[\omega_p(t - \tau) + \phi_p] * \cos[q * \omega_{LO}(t - \tau) + \theta_q] \\
&= \sum_q \exp^{j[\omega_p(t-\tau)+\phi_p-q\omega_{LO}(t-\tau)-\theta_q]} \\
&\quad + e^{j[-\omega_p(t-\tau)-\phi_p+q\omega_{LO}(t-\tau)+\theta_q]} + \dots
\end{aligned} \tag{2.1}$$

where, ω_p , ϕ_p , ω_{LO} and θ_q are the tones of the injected multisine signal, the associated phases, the harmonics of the local oscillator frequency and the phases of the harmonics of local oscillator frequencies, respectively. Note that the signal generator used and the local oscillator are synchronized by an external 10 MHz reference source the relationship between their phases will remain constant through out the measurement. Finally we can add the phase contribution of the IF filter at both positive and negative ω_p frequencies for the tones located within the IF bandwidth,

For positive ω_p frequencies :

$$\chi_p^+ = -\Delta\omega_p\tau + \phi_p - \theta_k + \varphi_{IF}[\Delta\omega_p] \tag{2.2}$$

For negative ω_p frequencies :

$$\chi_p^- = \omega_{LO}\tau - \Delta\omega_p\tau + \phi_p - \theta_{k+1} - \varphi_{IF}[\omega_{LO} - \Delta\omega_p] \tag{2.3}$$

If we expand these equations in order to include the effect of measurement time difference,

For positive ω_p frequencies :

$$\psi_p^+ = \Delta\omega_p t + \chi_p^+ \quad (2.4)$$

For negative ω_p frequencies :

$$\psi_p^- = -(\omega_{LO} - \Delta\omega_p)t - \chi_p^- \quad (2.5)$$

Due to the measurement time delay t , the measured phases for both the positive and negative frequencies will vary at each measurement. To solve this problem, two solutions are possible. The first one is to determine the greatest common divider frequency f_{gcd} for all then IF tones. Then this greatest common divider frequency f_{gcd} or a sub-harmonic of it is used for triggering the LSNA measurements. Under such conditions the relative phase acquired will become invariant. This approach is similar to the approach used in [6]. The other solution is to use an known input signal (for example an HPR) as a reference signal and extract the relative group signal and LO delay to compare successive measurements. This is indeed needed because τ is a function of relative measurement time t . Therefore, if τ is determined, one can find the relative measurement time delay. We have develop both methods for broadband phase calibration. However the only results for the trigger method will be shown in this chapter.

2.2 Measurement Set-up Description

A broadband measurement system integrated with a LSNA is presented in Fig. 2.1. An Agilent vector signal generator (E4438C) is used for the IF calibrations at the RF harmonics

and as a signal source is used for generating the modulated RF signal exciting the DUT at the fundamental frequencies. A conventional LSNA RF calibration including SOLT, power meter, and harmonic phase reference is used for the RF fundamental and harmonics. Then additional IF calibration procedures are performed at each harmonic band of interest. The IF calibration at each harmonic include an IF calibration of the incident signal at port 2 and a thru calibration between port 1 and 2. For the IF calibrations a signal of 63 tones with equal amplitude and specific phases are programmed in the vector signal generator for each of the harmonic of interest. To compensate for potential power differences between the programmed and measured signal at port 2, a spectrum analyzer or a vector signal analyzer can be used to calibrate the power level and phase of each tone, respectively. Finally, the thru and IF calibration at each harmonic are used to remove any differential dispersion in the IF channels of a_1 and b_2 between port 2 and 1.

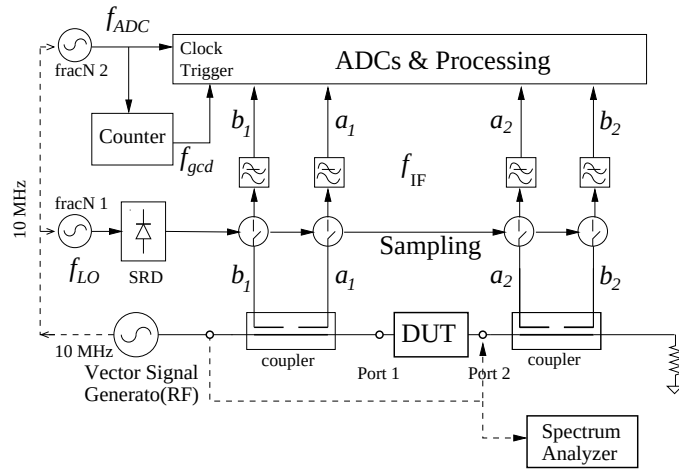


Figure 2.1: Block diagram of the modified LSNA setup and integrated IF calibration set-up.

Note that the measurement environments used at calibration is kept during the DUT measurements in order to maintain the calibration conditions. As the source power level changes, the test-set attenuation level is selected such that the LSNA keeps operating in its linear operation region during the IF calibration and the broadband measurements while maintaining a good dynamic range. Correction to the calibration algorithm is then performed as the attenuation level is changed in the testset.

2.3 Experimental Results and Discussions

The phases of a_1 and b_2 at the 63 FFT bins of interest for a 63 tones 80 MHz bandwidth signal measured with the broadband testbed are shown in Fig. 2.2 for a thru after RF and IF calibration corrections have been applied. Thanks to the properly locked measurement 1 Hz trigger, the phases of a_1 and b_2 at corrected bins remain the same in three successive measurements and correspond to the phases programmed in the vector signal generator. Note that identical phases between a_1 and b_2 results from the fact that a thru is measured. Fig. 2.3 illustrates the phases difference observed among three successive measurements. The maximum error range among measurements at each bin, indicates (1) that our calibration method is reliable, (2) that the trigger generated effectively synchronizes the measurement system and (3) that the vector signal generator generates a multitone RF signal with stable phases.

Fig. 2.4 gives the results obtained for a 480W GaN HEMT Doherty PA. Displayed in Fig. 2.4 and 2.5 are the spectra and phases for the periodic signal at the input (blue) and output (red) of the PA for three successive measurements. The output spectra was shifted (by 37 dB) to more clearly reveal the the spectral regrowth generated by the PA. Note that the 480W GaN HEMT Doherty PA is biased using three DC power supplies, $V_{GS,carrier}$

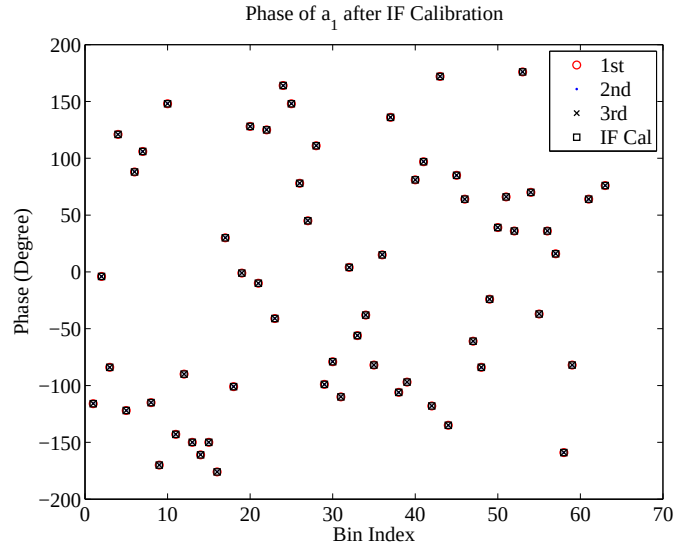


Figure 2.2: b_2 and a_1 phase after IF calibration for three successive measurements (blue is for a_1 and red is b_2).

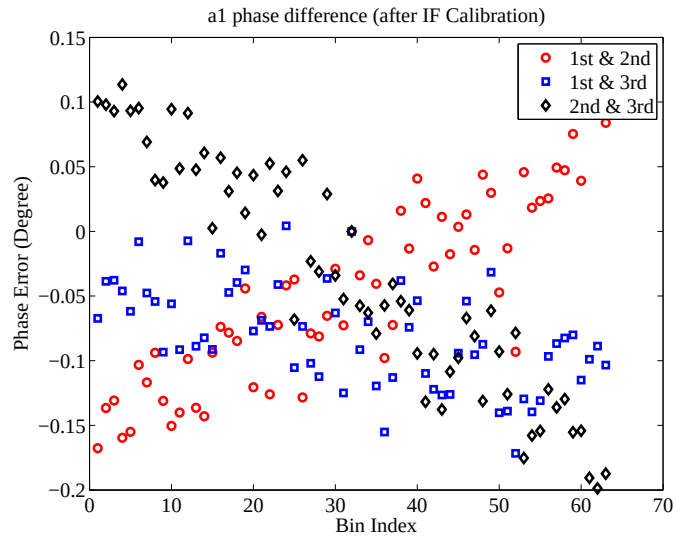


Figure 2.3: a_1 phase difference among three successive measurements.

of -3 V, $V_{GS,peaking}$ of -5 V and V_{DS} of 28 V, were selected for the DC biasing, yielding a drain current I_{DS} of 1 A and 0 A for the carrier and peaking amplifiers respectively. The fundamental frequency was set to 2.16GHz (a bit off centered relative to the PA Doherty center frequency) with a multi-tone bandwidth around 15 MHz. The number of tones is 13 tones and the phases are selected to yield a peak to average power ratio (PAPR) of 6 dB. The 80 MHz bandwidth used is wide enough to capture the 3rd and 5th order intermodulation bands. The input driving level was set to 2.7 dBm in the vector signal generator so as to yield a strong spectral regrowth in the PA. As seen in Fig. 2.4, the spectral regrowth of b_2 is more pronounced than of a_1 . Also, phases correspondence in a_1 among three measurements is shown around 15 MHz at center frequency. However, due to nonlinearity of PA, correspondence of phases in b_2 is presented whole frequency bandwidth. The spectral regrowth in the second harmonic of the PA is also presented in Fig. 2.6 for three successive measurements. The most reproducible measurements are obtained for tones above -10 dBm. Note that the procedure for the IF calibration used for the second harmonic data is similar to that used for the fundamental frequency.

Beside the spectral regrowth, the Error Vector Magnitude (EVM) is often used for qualifying the inband performance of a PA for a digital communication system. However, for obtaining highly reliable EVM, it is necessary that the measurement noise error in phase and amplitude contributed by the test system be smaller than those introduced by the PA. The EVM can also be used to quantify the vector measurement reproducibility of the test system. The reproducibility's EVM at the fundamental frequency for our testbed was found to be within 0.1 % whereas the input-to-output EVM of the PA was typically 11 % or above. These results bring additional confidence in the applicability of this broadband testbed for EVM measurement.

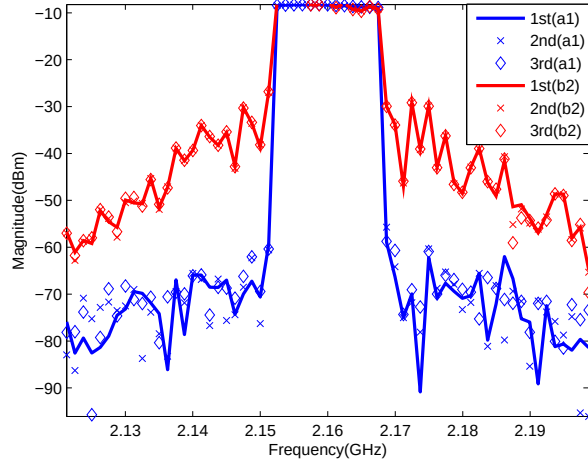


Figure 2.4: Spectral distribution of a_1 and b_2 at the input and output of the PA respectively.

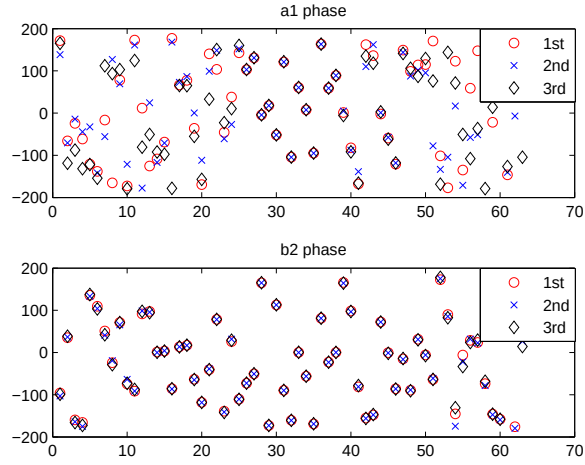


Figure 2.5: Phase distribution of a_1 and b_2 at the input and output of the PA respectively.

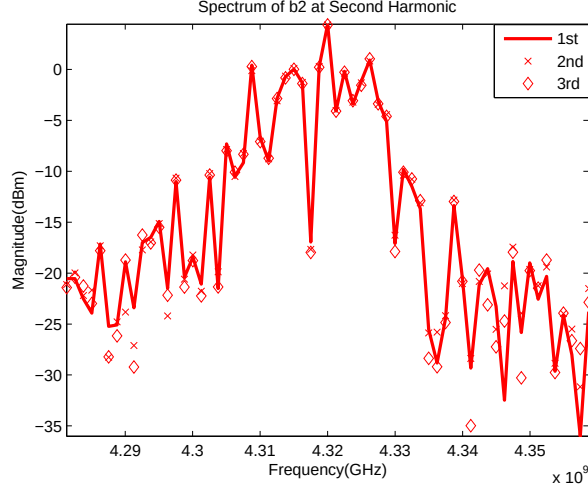


Figure 2.6: Resulting spectral regrowth in the 2^{nd} Harmonic at the PA output.

	Error Vector Magnitude between measurements
1st result& 2nd result	0.17 %
1st result& 3rd result	0.23 %
2nd result& 3rd result	0.14 %

Table 2.1: EVM Comparison between three successive measurements

2.4 Conclusion

A broadband measurement testbed was developed with an LSNA. This testbed is intended for characterizing PAs using periodically modulated (multitone) signals often referred as multisines. In the experiment conducted a multisine with 13 tones distributed over 15 MHz was used which provided a PAPR of 6 dB. Using this excitation, the non-linear response of a 480 W Doherty GaN PA at 2.16 GHz was characterized over a 80 MHz bandwidth. Inband signal distortion and spectral regrowth was recorded in the 63

tones measurement over the 80 MHz measurement bandwidth targeted at both the fundamental and the second harmonic. This vector measurement enables one to characterize the nonlinear response of the PA in its 3rd and 5th order inter-modulation bands at the fundamental frequency. The broadband vector results at the second harmonic obtained in the same measurements provides additional invaluable information for the multi-harmonic optimization of the efficiency of the Doherty PA under test. It is believed that the multi-harmonic broadband vector characterization provided by the reported testbed should assist with the development and verification of multi-harmonic behavioral models for wideband modulated signals.

CHAPTER 3

PULSED-RF REAL-TIME ACTIVE LOAD-PULL TECHNIQUE WITH A PASSIVATED ALGAN/GAN HEMT USING A LARGE SIGNAL NETWORK ANALYZER

As the complexity of commercial signals such as LTE is growing to address the increasing need for bandwidth and user capacity in wireless communication, the peak to average power ratio (PAPR) is also increasing reaching 10 dB for LTE. As the PAPR is increased, the operation point of power amplifiers (PA) must be proportionally backed off from their own 1 dB compression gain point in order to avoid hard non-linearities. Thus at peak power the PAs will be operating at their 1dB compression while keeping an acceptable signal distortion.

It results that the 1 dB compression gain point remains of great importance in the evaluation of the transistor linearity and the design of power amplifiers [7]. To determine the 1dB compression for a fixed load impedance a power sweep measurement (multiple measurements) is required. Considering that at least hundreds to thousands different impedance points should be measured to generate the P1dB load impedance contour plot, this is usually a very measurement intensive process. A similar large number of measurements is also required if the maximum power efficiency impedance termination is targeted for different output powers. Clearly the measurements required to determine the optimal load conditions for maximum output power under linear or power-efficient operation constrains are

quite time consuming and this greatly impacts the PA design cycle. A time efficient scheme would therefore be greatly beneficial considering that ideally P1dB load impedance contour plot measurements should be available for multiple biasing conditions or multiple modes of operation (harmonic termination).

Passive load-pull using mechanical tuners is the most common technique used to find the optimal load impedance contours yielding the maximum output power and efficiency in transistors. However, passive load-pull systems suffer from several limitations. First of all, measurement with passive load pull systems is typically a very time-consuming process due to the use of mechanical tuners. Second, passive tuners are limited in term of the maximum reflection coefficients they can attain due to the tuners' inherent loss [8]. To address this issue, active load pull (ALP) systems based on the closed-loop architecture [9], [10] have been developed to improve the reflection coefficient attainable. However, due to potential stability problem, a complex controlling scheme is required and closed-loop ALP systems are not extensively used. The stability issue was eliminated with the development of the open-loop ALP architecture [11]. However the long measurement time issue still remains an issue. This has been resolved with the advent of the real-time active load-pull (RTALP) measurement technique [12], which is possible to address all the issues. RTALP can dramatically reduce the required measurement time and there is no limitation in the attainable reflection coefficient without the risk of stability issue. This approach is thus considered to be highly promising to facilitate the design of linear power-efficient PAs when used in combination with nonlinear vector network analyzer (NVNA).

For successful device modeling and characterization it is also essential to account for slow memory effects such as self-heating, traps in HEMTs and parasitic bipolar transistor in

SOI or SOS MOSFETs which are originating from various physical processes in the transistors [13]. Due to the slow time response of these memory effects, fast pulsed-IV/pulsed-RF measurements have been used to maintain constant device temperature, body BJT voltage in FETs or trap state in HEMTs [14]. Several studies using pulsed RF measurements have been reported. A pulsed VNA was developed for a pulsed ALP measurement configuration to detect the pulsed-RF power within a given pulse width [15]. A pulsed load-pull system using a digital scope and a power meter was also introduced to measure the power gain of devices under pulsed condition [16]. In addition, a pulsed ALP system with an LSNA was used to design high-efficiency Class-B RF PAs with AlGaIn/GaN HEMTs in [17]. The previous studies in the pulse measurement system, however, were subject to the boundary of ALP configuration, which still require long measurement times in order to search for an appropriate load condition for the targeted applications. Moreover, it is well known that pulse measurements with VNAs suffer from a significant loss of dynamic range as the duty cycle of the pulse decreases. This is characterized by the desensitization factor:

$$\text{Desensitization} = \frac{P_{peak}}{P_{fund}} = 20 \times \log(\text{Duty_rate}). \quad (3.1)$$

where P_{peak} is the peak instantaneous RF power of the signal measured and P_{fund} the average RF power at the fundamental measured by the test equipment. To compensate for the loss of dynamic range, a modified architecture was introduced with the LSNA, in which all of the tones within the main lobe of the pulse signal are acquired to keep 90 % of the power [14]. The desensitization is then reduced to

$$\text{Desensitization} = 10 \times \log(\text{Duty_rate}). \quad (3.2)$$

Nevertheless, this approach requires some signal processing and is still restricted to duty cycle above 0.3 %. Indeed the total RF power is spread into multiple tones of amplitude

proportional to the desensitization and eventually the power of each tone will be fall below the minimum detectable level of the system as the duty cycle decreases. Recently a new approach, the multiple recording technique, was demonstrated for the LSNA which enables one to measure a pulsed signal for arbitrarily low duty cycles without any dynamic range reduction [18], [19]. There is thus no desensitization associated with multiple recording:

$$\text{Desensitization} = 0 \text{ dB.} \quad (3.3)$$

Indeed in this technique, a pulse control board inside the LSNA controls the analog digital converter(ADC) acquisition so that the sampled down-converted RF data are only acquired when the peak RF pulse is applied to the DUT. It can be experimentally verified that the data acquisition in the modified LSNA does not exhibit any degradation in dynamic range as the pulse duty cycle decreases [20]. The limitation in the pulse-width is only limited itself by the latency of the low-pass filter used after the sampler in the LSNA.

In this work, the RTALP and the multiple recording techniques were combined in order to reduce the exhaustive measurement time associated with the load-pull as well as to solve the desensitization problems associated with the pulse excitation. With this combined technique, the P1dB compression gain extraction was conducted to find the optimal load condition for the output power and gain on the entire Smith Chart. For the sake of verification of the new technique proposed in this paper, a passivated AlGaIn/GaN HEMT was selected as a device-under-test (DUT). Silicon nitride(SiN) passivation and field plates were used in this device to minimize the unwanted memory effects on the HEMT devices. The selected DUT is therefore expected to yield fairly similar measurement results in both continuous wave (CW) and pulsed excitations. Furthermore, the measurement results obtained under the two different measurement schemes (RTALP, Constant Phase) were compared for three

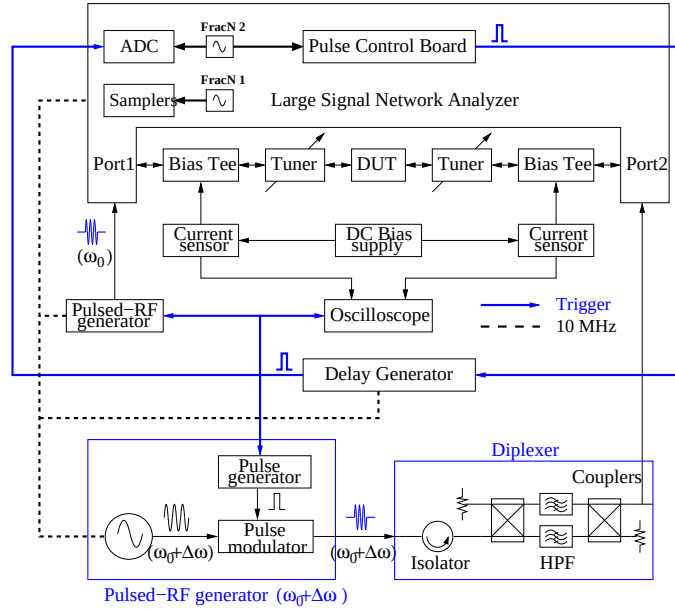


Figure 3.1: Pulsed real-time active load-pull system configuration

critical load terminations: maximum output power, maximum gain, and matched load (50 ohms).

It is to be noted that for devices exhibiting memory effects, pulsed-RF RTALP has the additional benefit of eliminating the memory effects potentially induced by the real-time sweeping of the RF phase (typically on the order of 10-100 kHz) taking place in CW RTALP.

The pulsed RTALP measurement configuration with the modified LSNA and DUT are explained in Section 3.1. Section 3.2 describes the methodology for the 1 dB compression gain plane extraction and presents the output power and gain results achieved at the 1 dB gain compression point. Finally, the conclusion is presented in Section 6.4.

3.1 Pulsed RF Real-Time Active Load-Pull System

Fig. 3.1 illustrates the configuration of the pulsed RTALP measurement system developed. The top and bottom signal generators, a pulse generator and a modulator are combined to make and then drive pulsed RF signals at ω_0 and $\omega_0 + \Delta\omega$ into both ports for the real-time active tuning. The DC bias combined at the bias tee is then fed to the DUT.

The pulse control board in the LSNA controls the master trigger feeding the delay generator which establishes the synchronization between the pulsed-RF sources, the pulse modulator, and the data acquisition performed by the ADC of the LSNA. In addition, the pulsed-RF source provides the pulse control board, the first $\frac{1}{N}$ for samplers, the second $\frac{1}{N}$ for the ADC, and the delay generator with the 10-MHz reference clock signal in order to share the same time base.

A passivated AlGaIn/GaN HEMT with a 160 nm SiN layer on the surface was used for testing the designed pulsed-RF RTALP system. The device was grown on SiC substrate and composed of a 830 nm thick unintentionally doped (UID) GaN layer which lies on a 50 nm thick AlN layer upon a 30 nm thick AlGaIn layer with 28.3 % Al composition [20]. The gate length and width are 0.7 μm and 150 μm , respectively.

3.2 Experimental Results

In this section we shall discuss the measurement results obtained. The two major goals of this measurement are 1), to discover the optimal load impedances where the maximum output power and gain are obtained in linear gain region under pulsed-RF condition, and then 2) to determine whether the developed pulsed-RF RTALP system is reliable by comparing to the results obtained with those obtained under CW excitations. In general, the

output power and gain between CW and pulsed-RF RTALP measurements will be different due to self-heating and trapping effects triggered under CW excitation. However, if the passivated AlGaIn/GaN HEMTs is not strongly affected by slow memory effects the RTALP measurements in both conditions will yield similar results.

Furthermore, the results obtained from RTALP measurements will be compared to those obtained from constant phase(ALP) measurements, specifically at three critical impedance points corresponding to the maximum output power, the maximum gain, and the matched load condition (50 ohm). This will help to delineate the impact of the memory effects associated with the real-time sweeping of the phase in CW ALP. In all measurements, the fundamental frequency is 2 GHz and the modulation frequency (Δf) which is used for the RTALP measurements is 200 kHz. Four harmonics are also considered in the measurements. The quiescent bias point for the passivated AlGaIn/GaN HEMT transistor used is $V_{GS} = -2.00$ V, $V_{DS} = 5.00$ V, and $I_{DS} = 45.0$ mA. Also, note that a pulse width of 10 μ s with a duty rate of 1 % was used for the pulsed-RF signals in all these measurements given the latency limitation of the particular LSNA receiver used.

3.2.1 P1dB Reference Incident Power for CW and pulsed-RF conditions

Prior to extracting the 1 dB compression gain plane on the Smith Chart, it should be first verified that the selected low incident power (here -5 dBm) allows all load impedances on the Smith Chart to remain in the linear gain region. In both of the testbeds used for the CW or pulsed-RF measurements, the pulsed-RF signal generator (MG3692A) is used to drive the fundamental frequency (ω_0) into port 1, and the combined ESG signal generator and pulse modulator is used to inject the $\omega_0 + \Delta\omega$ excitation to port 2 for the real-time tuning. In Fig. 3.2, the red solid and the blue dashed lines are for the $\Gamma_L(\omega_0, t)$ loci

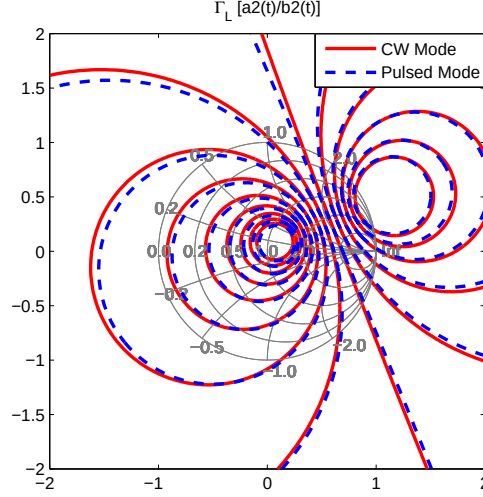


Figure 3.2: Loci of $\Gamma_L(\omega_0, t)$ obtained from the $(\omega_0 + \Delta\omega)$ real-time swept phase measurements in CW (red solid line) and pulsed (blue dashed line) mode at input power of -5 dBm.

obtained in both CW and pulsed-RF conditions respectively. The two types of line are nearly superposed on top of each other and cannot be easily differentiated. Three dominant tones $(\omega_0 - \Delta\omega, \omega_0, \omega_0 + \Delta\omega)$ were considered for the calculation of $\Gamma_L(\omega_0, t)$ loci.

It is evident that a sufficiently dense set of load reflection coefficients on Smith chart are mapped in these measurements.

Fig. 3.3 shows the results for the output power (top) and the gain (bottom) contours obtained from RTALP measurements at -5 dBm of incident power. The top plot of the Fig. 3.3 compares the output power contours obtained from the CW (solid contours) and the pulsed-RF excitation (dashed contours), and the bottom plot of the Fig. 3.3 compares the power gain contours obtained from the CW (solid contours) and the pulsed-RF conditions (dashed contours). The power gain is defined as the ratio of P_{in} over P_{out} . The unit used for the gain contours in this paper is the direct power ratio instead of the dB scale.

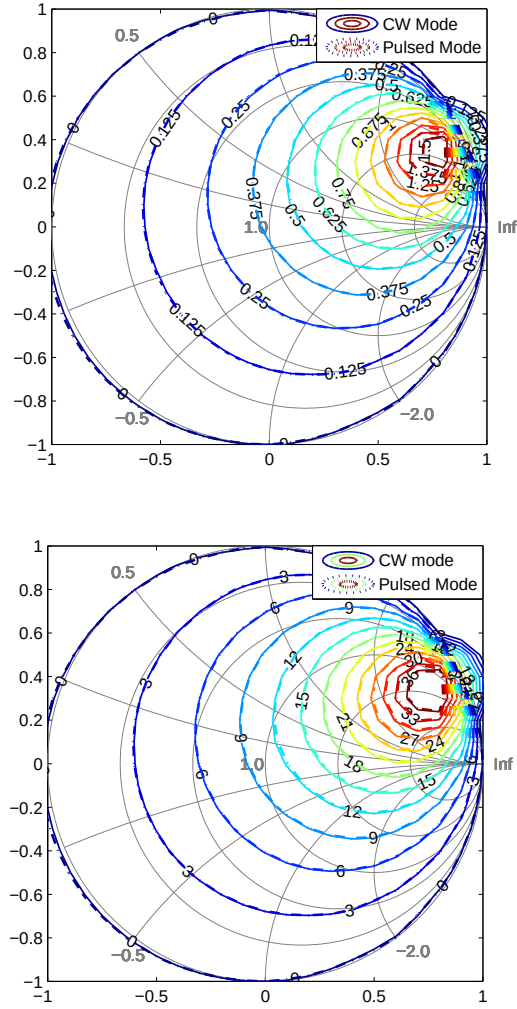


Figure 3.3: Output power (top) and gain (bottom) contours obtained from the $(\omega_0 + \Delta\omega)$ real-time swept phase measurements in CW (solid line) and pulsed (dashed line) mode at input power of -5 dBm.

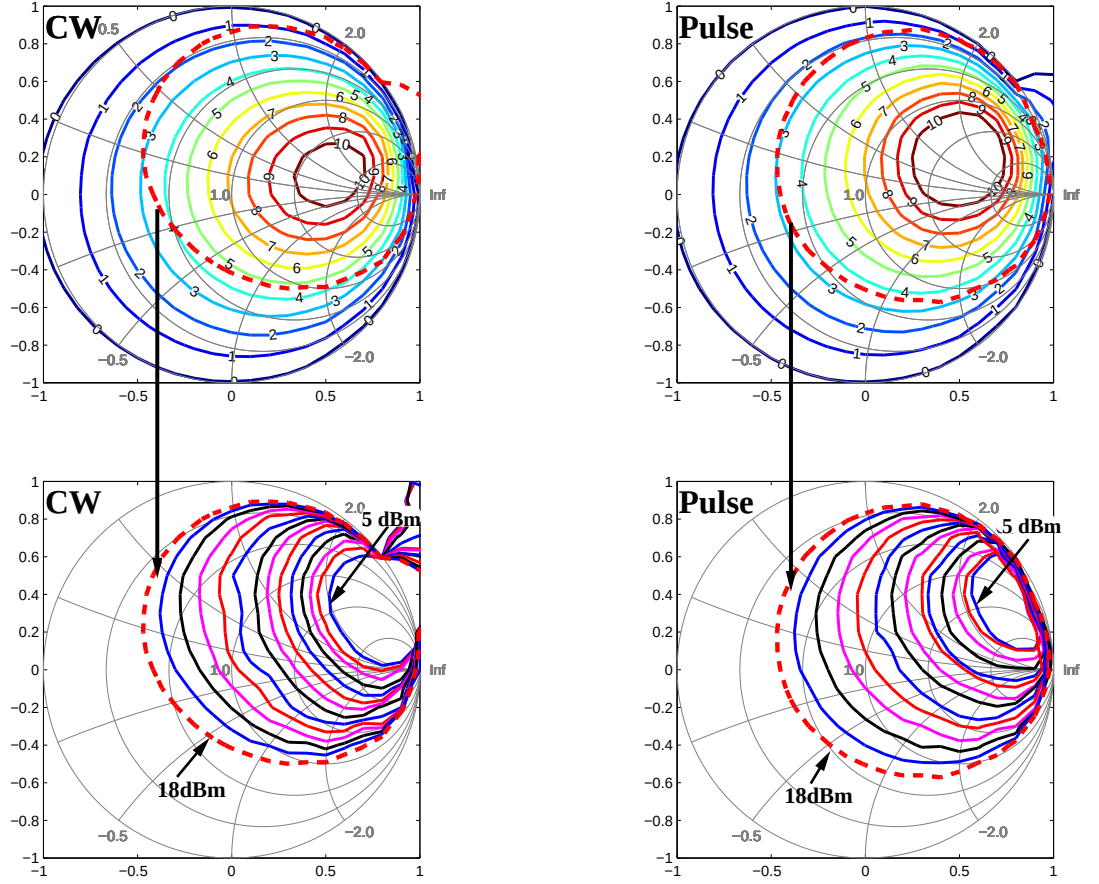


Figure 3.4: The top figures show the gain contours obtained from the real-time active load-pull measurements at incident power of 18 dBm in both CW and pulse modes. The red dashed line in top and bottom plots show the locus of the corresponding 1 dB compression gain at 18 dBm incident power. The bottom plots show the cumulation of the extracted 1 dB compression gain loci from 5 dBm to 18 dBm in both CW and pulse modes.

The maximum gain and output power of 16.2 dB, 1.3 mW and 16.2 dB, 1.6 mW, respectively, are obtained at the same load impedance $\Gamma_L(\omega_0) = 0.8+j0.4 (=0.89\angle 26.6)$ in CW and pulse modes. However, this load impedance is located in a potentially unstable region due to the negative resistance exhibited at the input of the transistor. In two-port

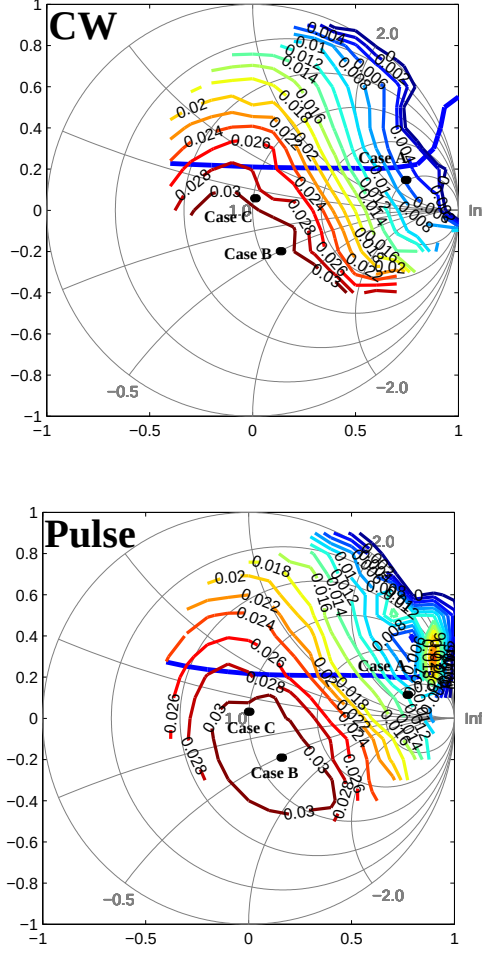


Figure 3.5: Output power contours on 1dB compression gain plane obtained in CW mode (left) and pulse mode (right) with $|\Gamma_{in}(\omega_0, t)| = 1$ from the real-time active load-pull measurements

network, oscillations may occur when $|\Gamma_{in}(\omega_0)| > 1$ or $|\Gamma_{out}(\omega_0)| > 1$. With this respect, the areas with positive input impedances ($|\Gamma_{in}(\omega_0)| < 1$) are only investigated in the rest of this paper. Thus, an alternative load impedance for maximum gain could be $\Gamma_L(\omega_0) = 0.7+j0.2$ ($=0.73\angle 15.9$) in both conditions and the corresponding gain and output power are 15.24 dB, 1.29 mW and 15.20 dB, 1.28 mW, respectively.

Table 3.1: Comparison of output power and gain in CW and pulse mode

$\Gamma_L(\omega_0)$	CW		Pulse	
	$P_{out}(mW)$	Gain(dB)	$P_{out}(mW)$	Gain(dB)
0.7/15.9:	1.29	15.24	1.28	15.20
50ohm	0.40	10.17	0.40	10.12

Based on the contour plots, Table 3.1 shows a comparison of the output power and gain in both conditions at the obtained optimal load impedance and 50 ohms at ω_0 when the incident power is fixed at -5dBm. The difference is only 0.01 mW in the Max. output power for $\Gamma_L(\omega_0)$ and 0.05 dB in the gain for 50 ohms. Thanks to the low incident power selected as well as the low memory effects exhibited by the device itself, the memory effects were not activated in the CW excitation so that the results obtained for both CW and pulse conditions are quite similar.

3.2.2 The Extraction of 1 dB Compression Gain Plane

In the previous section, the gain and output power contours at -5 dBm of incident power were selected as the reference for the 1 dB compression gain plane extraction(see Fig.3.3). In Fig. 3.4, the top two figures illustrate the gain contours at the incident power of 18 dBm for the CW (left) and pulsed-RF (right) excitations respectively. Also superposed on the same top figures is a red dashed line corresponding to the 1 dB compression gain locus extracted relatively to the low-power gain obtained for -5 dBm incident power in Fig.3.3. The bottom two figures show the cumulated set of 1 dB compression gain obtained relatively to -5 dBm low-power gain contours of Fig.3.3 under both CW and pulsed conditions when the incident power is swept from 5 to 18 dBm by 1 dB step. The 1 dB compression

Table 3.2: The Output power and Gain Comparison of Three Typical Cases at P1 dB points: Case A, B, C by RTALP

$\Gamma_L(\omega_0)$	$P_{out}(mW)$		$Gain(dB)$	
	<i>CW</i>	<i>Pulse</i>	<i>CW</i>	<i>Pulse</i>
Case A : (Max. Gain 0.7/15.9)	6.0	10.5	14.9	14.9
Case B : (Max. P_{out} 0.3/-71.6)	35.0	32.2	8.4	8.4
Case C : (50 Ohm)	27.1	30.4	10.8	11.6

gain curve (red dashed line) extracted in the top figures for 18 dBm incident power is one member of this set of curves.

3.2.3 Output Power and Gain Comparison with CW and pulsed modes on the 1 dB Compression Gain Plane

The subset of measured power gain and power data available along the 1dB gain compression loci extracted in Fig. 3.4 provides the final database for the power gain and output power targeted for the P1dB compression plane. Fig. 3.5 and 3.6 illustrate the output power and power gain contours along this 1 dB compression gain plane, obtained for CW (left) and pulsed-RF (right) excitations, respectively. In these figures, the $|\Gamma_{in}(\omega_0)|$ boundary is also depicted by a thick blue (quasi-horizontal) line. The extraction of $\Gamma_{in}(\omega_0)$ is also conducted at the same time at that of the output powers and power gains.

Output power and gain at three critical load conditions corresponding to the maximum output power and maximum gain, as well as the standard 50 ohms matched load are summarized in Table 3.2. For the Case A and C load conditions, a little bit higher output power

Table 3.3: Averaged Output power and Gain Comparison of Three Typical Cases at P1 dB points: Case A, B, C by RTALP

$\Gamma_L(\omega_0)$	RTALP			
	$P_{out}(mW)$		Gain(dB)	
	<i>CW</i>	<i>Pulse</i>	<i>CW</i>	<i>Pulse</i>
Case A : (Max. Gain 0.7/15.9)	7.0±5.3	8.0±2.9	14.9±0.0	15.0±0.1
Case B : (Max. P_{out} 0.3/-71.6)	36.4±1.4	32.4±1.0	8.4±0.0	8.4±0.0
Case C : (50 Ohm)	27.7±1.0	29.2±2.0	10.7±0.1	10.7±0.0

Table 3.4: Averaged Output power and Gain Comparison of Three Typical Cases at P1 dB points: Case A, B, C by Constant Phase

$\Gamma_L(\omega_0)$	Constant Phase			
	$P_{out}(mW)$		Gain(dB)	
	<i>CW</i>	<i>Pulse</i>	<i>CW</i>	<i>Pulse</i>
Case A : (Max. Gain 0.7/15.9)	10.7±0.3	11.5±0.3	14.3±0.2	14.6±0.1
Case B : (Max. P_{out} 0.3/-71.6)	26.7±0.7	27.9±0.7	8.8±0.1	8.9±0.1
Case C : (50 Ohm)	26.1±0.3	26.7±0.2	10.7±0.1	10.7±0.1

and gain are measured under pulsed-RF condition than CW-RF operation even though these difference is within quite small range. Possibly, the difference are caused by the inherent low-frequency dispersion arising from AlGaIn/GaN HEMTs at the selected loads. However, with the selected load in Case B, one can find more output power in CW than in pulse mode which is in contradiction with Case A and C. This is most likely due to the fact that the Case B extraction is achieved at the high incident power of 18 dBm with the transistor operating under continuous Class-A bias operation. Therefore, under such conditions the temperature of the DUT will be decreased in CW-RF compared to the pulsed-RF thanks to the high DC to RF power conversion. Indeed as noted before in the pulsed-RF measurement, the DC operating point was not simultaneously pulsed.

To evaluate the memory effects caused by the real-time tuning of the loads, Table. 3.4 compares the averaged output power and gain and their fluctuation in three successive measurements under CW and pulsed conditions for two different experiment methodologies: RTALP and Constant phase measurement. Note that, in the constant phase measurement, the ESG source injects the 2 GHz fundamental signal at the port 2 without any frequency offset.

The difference in output power for Case B may be in part due to the difference in selected incident power between the constant phase and RTALP measurements. Due to extraction range resolution (1 dB), the selected incident power (17.0 dBm) for the constant phase (ALP measurement is possibly lower by 0.5 dB than the incident power (17.5 dBm) for RTALP measurement.

Having taking this effect into account a lower variation in memory effects associated with the phase-sweeping in RTALP is expected when using pulsed-RF excitation versus CW excitation. This is verified for case B, where a relatively smaller variation of the

output power of 4.5 mW (0.65 dB) for pulsed-RF excitation compared to 9.7 mW (1.35dB) for CW operation when comparing the RTALP and ALP measurements.

For case A, a relatively high fluctuation of 5.3 dB fluctuation in output power is obtained compared to the smaller ones observed in case B and C. One possible explanation for this effect is that the load location of case A is in the vicinity of the $|\Gamma_{in}(\omega_0)| = 1$ boundary (see Fig. 3.7) and is thus close to the potentially unstable region ($|\Gamma_{in}(\omega_0)| > 1$).

In summary the comparison of RTALP and ALP for CW and pulsed-RF excitations confirms that residual memory effects at high incident RF power are reduced under pulsed-RF operation compared to CW operation.

3.3 Conclusion

A pulsed-RF RTALP measurement system was implemented with an LSNA and the results obtained were compared to CW RTALP and constant-phase ALP measurements. The recently reported multiple recording technique was used for the acquisition of pulsed-RF signals to perform the pulsed-RF RTALP measurements without any reduction in dynamic range. This testbed which greatly reduces the measurement time and operator effort compared to the conventional passive methods was then utilized to generate the constant output power at 1dB gain compression over the entire Smith chart.

The experiments were performed with a passivated AlGaIn/GaN HEMT transistor operating in class A. To reduce the complexity of the measurement and not add additional sources of uncertainties in this testbed evaluation, pulsed-biasing was not activate during the pulsed-RF measurements. The optimal load terminations for maximum power gain and maximum output power in the pulsed excitation were determined for this constant bias class A operation. The results obtained were then compared to those obtained for CW excitation

for the purpose of verifying the performance of the new pulsed-RF RTALP system. Also, constant phase measurements at three critical load points were acquired to further verify that the memory effects caused by the real-time tuning were reduced in pulsed-RF RTALP compared to CW RTALP.

This new methodology provides great potential for speeding up the characterization of non-linear power amplifier design while removing memory effects including self-heating and trapping. This pulsed-RF load-pull testbed should find application in the design of RF amplifiers intended for communication signals exhibiting high PAPR for which the device temperature is determined by the average output power and not the peak output power.

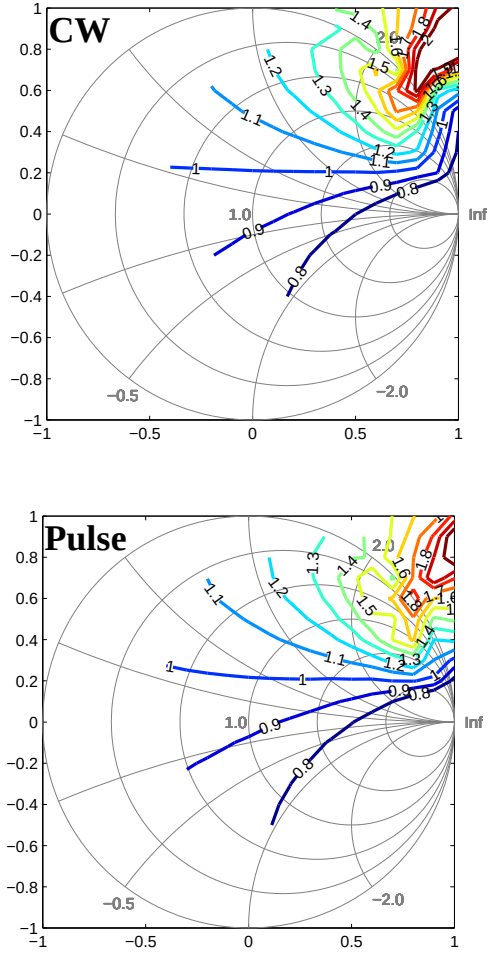


Figure 3.7: $|\Gamma_{in}(\omega_0)|$ contours of 1dB compression gain plane obtained from power sweep at $(\omega_0 + \Delta\omega)$ in CW mode (right) and pulse mode (left)

CHAPTER 4

NEW THERMOMETRY AND TRAP RELAXATION CHARACTERIZATION TECHNIQUES FOR ALGAN/GAN HEMTS USING PULSED-RF EXCITATIONS

AlGaN/GaN HEMTs have demonstrated impressive performance for high power RF amplification [21] [22]. However they can exhibit temporary current collapse or IV knee walkout which degrade their RF performance [23] [24] [25]. Many studies have been reported which ascribe such performance reductions to various memory effects such as self-heating, surface states and bulk traps [22]. Although many solutions have been devised to mitigate these memory effects, important RF dispersion effects can still be observed in Al-GaN/GaN HEMTs particularly under high RF voltage operation. The proper understanding and characterization of these memory effects is thus of critical importance for modern digital communication systems which feature modulated RF signals with high peak-to-average power-ratio (PAPR). New large-signal pulsed-RF measurement-techniques are presented in this chapter to address this issue.

4.1 Experimental Results and Discussions

The setup used for the pulsed-IV/RF measurements is shown in Fig. 4.1. New multiple recording data acquisition techniques were implemented in (1) the LSNA for the pulsed-RF

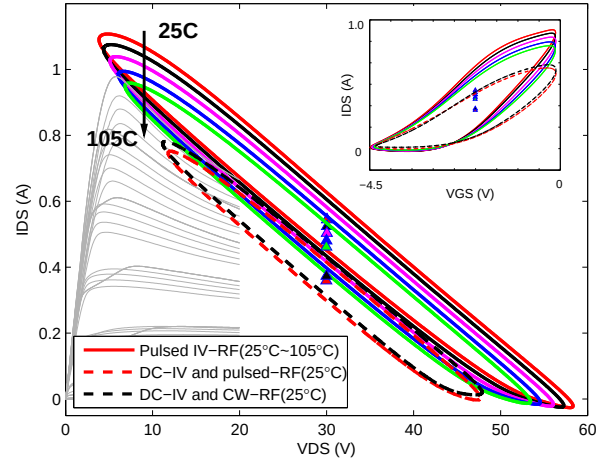


Figure 4.2: Comparison of the dynamic load-lines at 1 GHz obtained from pulsed-IV/RF measurements at various substrate temperatures (25 °C - 105 °C) and from CW-RF measurement at 25 °C. The same RF gate voltage swing (see inset) is applied at the input in all cases.

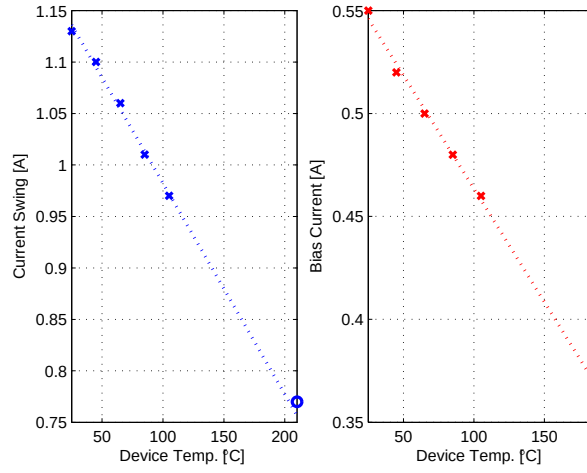


Figure 4.3: Variation of the pulsed-RF and bias drain current versus temperature and interpolated device temperature ($\approx 204^{\circ}\text{C}$: circles) and ($\approx 185^{\circ}\text{C}$: circles) .

dynamic load-lines measured for the CW measurements. The effective device temperature in DC-CW operation can then be estimated by extrapolation to be around 194 °C from both the RF drain current swing and bias drain current variation.

4.1.2 Trap Relaxation Characterization Techniques

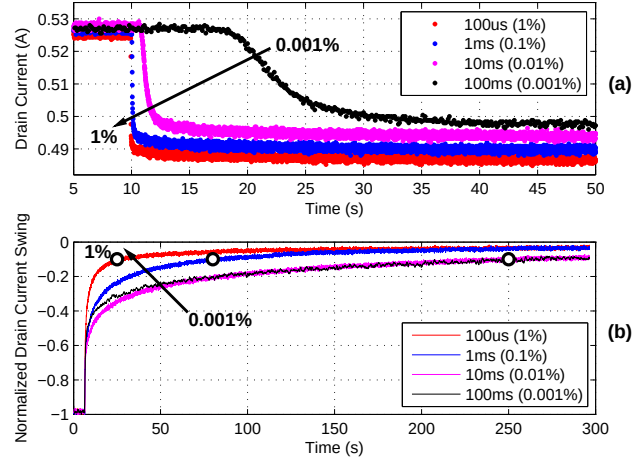


Figure 4.4: Transient bias drain current measurements for trap capturing (a) and emission (b) process under different duty rates. A normalized scale is used for the emission to better compared the relaxation times.

The emission and capture times of traps at various temperatures can be obtained by monitoring the long time relaxation of the transient bias drain current in pulsed-IV/RF measurements. Fig. 4.4 (a) and (b) present the slow relaxation versus time of the transient bias drain current measured by pulsed-IV measurements. When a train of pulsed-RF excitations with different duty rates is applied in Fig. 4.4 (a), the transient bias drain currents are found to decrease with time before settling down to a new stationary value. When the

pulsed-RF excitations are turned off in Fig. 4.4 (b), the transient bias drain currents are slowly increasing and relaxing to its original steady-state value.

The first process in Fig. 4.4 (a) is associated with traps capturing electrons when high instantaneous drain voltages ($\simeq 50$ V) result. The capture time of the traps is seen to be considerably lengthened through the use of pulsed-RF excitations with low repetition frequencies corresponding to duty rates varying from 1% to 0.001%. A cumulative relaxation capture time of about $150 \mu\text{s}$ corresponding to 150 RF pulses is required to fill the traps in these measurements. A delay of 80 RF pulses corresponding to a trap latency of $80 \mu\text{s}$ is also observed for the traps to start impacting the drain current.

The relaxation of the bias drain current when the RF is turned off is also found to be slowed down at low duty rates in Fig. 4.4 (b). The bias current relaxation is associated with the emission of electrons by traps when the device is both off ($V_{DS}=0$ V) and on (pulsed-bias of $V_{DS}=30$ V and gate bias yielding a drain current of 0.53 A). The trap emission time constants of both biasing states can be estimated by analyzing the dependence of the total relaxation time upon the duty rate DR. By simultaneously fitting the total relaxation rate $1/\tau_{total}$ at 90% recovery of the bias drain current using $\text{DR}/\tau_{ON} + (1 - \text{DR})/\tau_{OFF}$, the emission time constants when the RF is turned off are extracted to be $\tau_{ON} = 0.3$ s when the device is biased on and $\tau_{OFF} = 160$ s when the device is off. Clearly the emission times when the RF excitations are removed are considerably longer than the capture time of $150 \mu\text{s}$ obtained when the RF excitations are applied.

To confirm that trapping activity is involved, the device was further illuminated with a monochromatic beam with photon energies varying from 1.5 eV to 2.4 eV while performing the above pulsed-IV/RF measurement sequences. As is seen in Fig. 4.5, a significant reduction in the emission time constants is observed when the photon energy is increased.

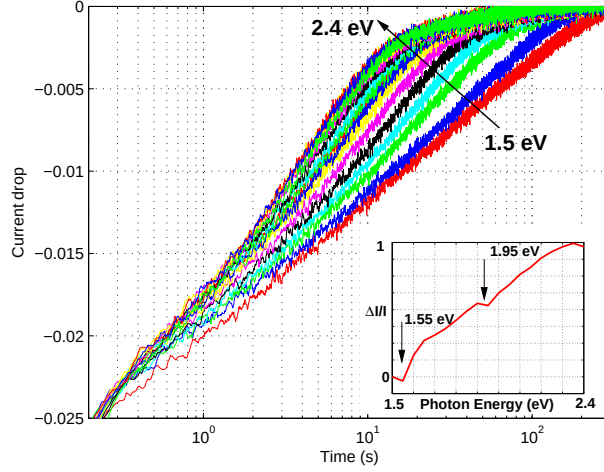


Figure 4.5: Transient bias drain current measurements for trap emission with the device subject to monochromatic illumination with photon energies from 1.5 eV to 2.4 eV. The inset shows the associated deep level spectroscopy under pulsed-IV/RF excitations.

The associated deep-level optical spectroscopy inset plot in Fig. 4.5 permits to identify that two deep trap energy levels located at 1.55 and 1.95 eV below the conduction band edge are involved in this process.

The negligible temperature dependence of the capture time which was observed for substrate temperature from 25 to 105°C suggests that the deep trap levels are populated in the presence of high instantaneous RF voltages by a tunneling process such as Fowler-Nordheim tunneling.

To further demonstrate the dependence of the trap capture on the instantaneous drain voltage applied, the pulsed-IV/RF measurements were conducted for two different load impedances of 10 and 50 Ω respectively while sweeping the input RF power. The RF loadlines (not shown) reveal that at maximum input RF power, the instantaneous drain voltage reaches 50 V for the 50 Ω load while it only reaches 37 V for the 10 Ω load.

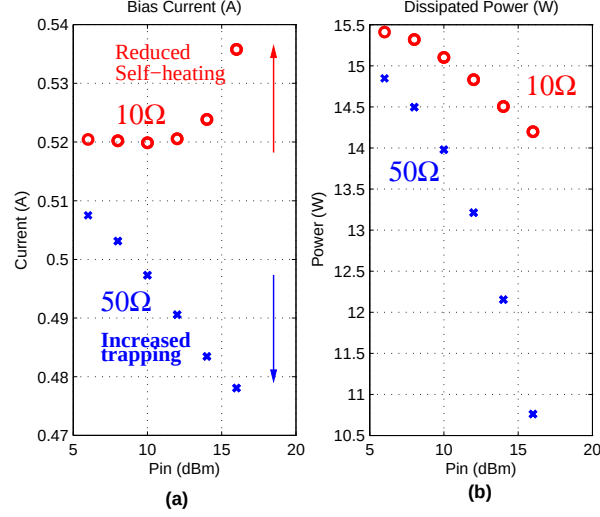


Figure 4.6: Comparison of the variation of bias drain current and the transistor dissipated power versus the RF input power for the $10\ \Omega$ (red circles) and $50\ \Omega$ (blue crosses) load impedances.

The resulting variation of the bias drain current versus the input power is shown in Fig. 4.6(a). Also shown in Fig. 4.6(b) is the variation of the dissipated power versus the input RF power. The power dissipated by the transistor operating in class A, decreases with increasing input RF power as the DC to RF conversion becomes more efficient.

The plots of the bias drain current shown in Fig. 4.6(a) for both loads enable to determine which of the self-heating and trapping memory-effect processes dominate. For the $10\ \Omega$ load case the traps are not activated and the red circles indicate an *increase* of the bias drain current with increasing input RF power due to the resulting reduced power dissipation and self-heating. For the $50\ \Omega$ load case, trapping dominates and the blue crosses indicate a *decrease* of the bias drain current due to the increase trap occupation with increasing peak instantaneous drain voltages at high input RF power. This decrease in bias drain current occurs despite the more efficient DC to RF conversion (reduced self-heating) observed for

the $50\ \Omega$ case compared to the $10\ \Omega$ case in Fig. 4.6(b) which would otherwise increase the drain current. These measurements for the different loads and input RF power clearly demonstrate (1) the critical role played by the high instantaneous RF voltage on the trap population and (2) the minor role played by the transient operating point.

4.2 Conclusion

In this chapter, it was demonstrated that the short capture time of the traps at high voltages can be considerably lengthened through the use of μs large-signal pulsed-RF excitations with duty rate as low as 0.001%. In addition, by comparing the device performance for two different load impedances, it was shown that (1) the average power dissipation determines the internal device temperature while (2) the instantaneous high drain voltage is responsible for the enhanced trapping.

CHAPTER 5

NONLINEAR MEASUREMENT AND CHARACTERIZATION OF RF TRANSISTORS WITH MEMORY EFFECTS

Various type of low-frequency memory effects arise in transistors due to different physical processes. Self-heating, trapping, and bipolar parasitic transistor effects in MOSFET are typically the main sources of low-frequency memory effects with electrical time constants running from 100 s to 100 ns. It is of interest to investigate the impact of these memory effects on the performance of power amplifiers excited by signals with high peak to average power ratio. This investigation of the dynamics of the memory processes taking place will provide us with insights in the practical requirements for accurately characterizing and modeling RF transistors.

Complex modulated signals such as WiMAX and LTE exhibit large peak to average power ratios (PAPR) reaching 10 dB. However PAPRs larger or equal to 10 dB arise with a very small probability (typically 10^{-5}) and take place for short duration on the order of 100 ns. Thus most of the time the amplifier is amplifying the modulated signals at the average power. Yet the amplifier still needs to amplify quasi-linearly the envelope peaks with a PARP up to 7 dB to maintain a reasonable bit error rate. Furthermore the amplifiers must be designed to operate with a good efficiency from the average power to the peak power to deliver acceptable average power efficiency. To successful complete this complex amplifier

design, the PA designers must definitively rely on simulation and measurement data which provide a realistic evaluation of the device performance.

Load-pull is an experimental technique which is often used either in simulation or experimentally to design power amplifiers. In load-pull measurements the multi-harmonic terminations presented by the load and source networks are tuned to optimize the power added efficiency (PAE) of the PA while maintaining its gain over the dynamic range of the output power. Conventionally load-pull relies on constant wave (CW) RF signals. However CW load-pull may not provide a reliable evaluation of the transistor performance if the device exhibits strong memory effects. To demonstrate this let us first consider the case of self-heating. For amplifiers operating in class B, AB or class F, the power dissipated by the transistor increases with increasing input power. However considering that (1) the peak power is only achieved for a very short duration (100 ns) and has a low probability of occurrence and that (2) the thermal effects are usually relatively slow (μs), the device will remain operating at the temperature established by the average dissipated power. Therefore CW load-pull at the peak power may not yield realistic results due to the comparatively much larger self-heating it will induce. Next consider the impact of traps in devices such as GaN HEMTs. Under low-power CW operation the drain voltage peak remain typically small and the traps are not activated. However for modulated RF signals, the instantaneous drain voltage reaches intermittently large values (50 to 60 V). Traps in GaN typically feature very short capture time (ns) at high drain voltages and very long emission time (s) at low drain voltages. It results that traps gradually charge during the brief the periods of peak power and do not have time to discharge during the intervening time. The dominant response of the transistor for the average input power is then directly

The impact of memory effects on the transistor performance excited by high PAPR can be detrimental (trap case) or beneficial (lower temperature of operation or reduced parasitic BJT) depending on the particular dominant memory process. Since CW load-pull measurements may not provide a realistic characterization approach, other alternatives must be pursued. To more realistically characterize the performance of transistor for RF signals with high PAPR modulation, pulsed RF large-signal measurements from well defined hot-bias operating condition offers such an alternative. The pulsed IV/RF testbed used in this section is shown in Fig. 5.1. The pulsed measurements with the LSNA relies on the method of multiple recording data acquisition [18], [40] to achieve 0% desensitization independently of the duty rate. The new technique of pulsed-RF real-time active load-pull (RTALP [41] [40]) measurements was used to optimize the output load.

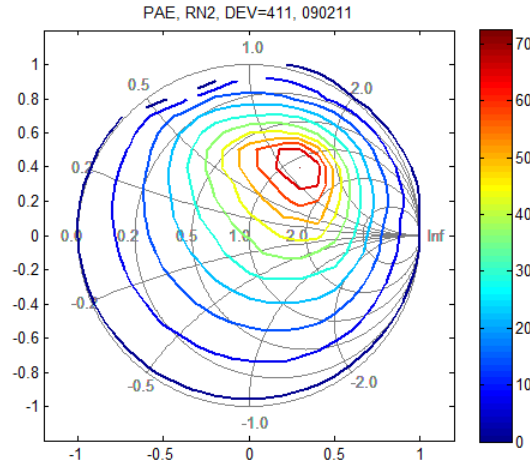


Figure 5.2: Constant PAE contour obtained from pulsed-RTALP shown here for 0.4 dBm incident power for class AB operation.

Table 5.1: CW & Pulsed RF Transistor Performance

Meas.	V_{GS} (V) (OFF/ON)	I_{DS} (mA) (OFF/ON)	P_{in} (mW)	P_{out} (mW)	Gain(dB)	PAE(%)
CW (1 dBm)	0.41	3/9.8	0.3	12.8	16.1	58.0
Pulsed (1 dBm)	0.41/0.41	3/10.0	0.3	14.5	16.7	65.9
CW (4 dBm)	0.41	3/11.7	0.6	16.7	14.6	63.0
Pulsed (4 dBm)	0.41/0.41	3/11.3	0.7	18.0	13.9	70.5
CW (11 dBm)	0.41	3/31.6	2.0	42.8	13.3	59.6
Pulsed (11 dBm)	0.41/0.41	3/33.1	2.5	54.3	13.4	73.7

To investigate the use of pulsed-RF large-signal measurements in the characterization of the memory effects we shall now consider a sub-cell of a power SOI-MOSFET transistor. This device will be initially tested for three different incident power levels of 1, 4 and 11 dBm, for CW and pulsed RF signals of 1% duty rate. The PAE contour plot obtained from pulsed-RTALP is shown in Fig. 5.2 for input power level of 0.4 dBm. The optimal load $\Gamma_L = 0.5\angle 37^\circ$ selected for low power operation yields a maximum PAE of 66%. Note that a different drain load of $\Gamma_L = 0.22\angle 117^\circ$ is used at 11 dBm incident power.

The RF transfer characteristic i_D versus v_{GS} and the dynamic load lines i_D versus v_{DS} obtained are shown in Fig. 5.3 and 5.4. The performance of the device is summarized in Table 5.1.

The gate voltage swing shown in Fig. 5.3 is seen to be about the same for the various incident signals. The output drain is also similar for the 1 and 4 dBm incident power levels. However for 11 dBm incident power the drain current swing is much larger. The drain current under pulsed RF reaches a more negative value indicating an increased power generation during the RF cycle associated with the drain charge storage. Fig. 5.4 also

shows a corresponding increased drain voltage swing. It is to be noted that the pulsed-RF load line reaches a lower knee voltage that expected from the DC and pulsed IV at $V_{GS}=2$ V (gray plain and dashed lines in Fig. 5.4). As indicated in Table 5.1, the pulsed RF measurements exhibit an improved PAE of 14% compared to the CW measurement while maintaining the same power gain.

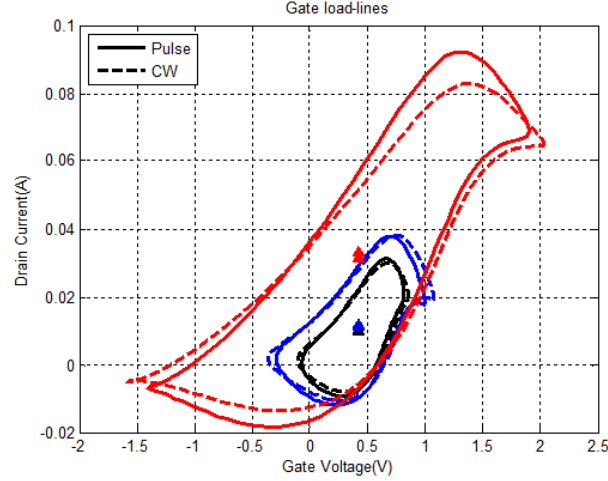


Figure 5.3: RF transfer obtained for CW (dashed lines) and pulsed (plain lines) RF excitations with incident power of 1, 4, and 11 dBm and 1 % duty rate.

To investigate the physical origin of this memory effect we shall now (1) vary the DC bias points of the pulsed-RF measurements or (2) change the duty rate. The results are shown in Fig. 5.5 and 5.6 and the transistor performance summarized in Table 5.2. The previous pulsed RF load line (red line with top triangles) which was measured with the device at the DC bias temperature (no RF applied) and the CW load line (black line with diamonds) are included for references. The pulsed RF load lines with a higher hot-biasing

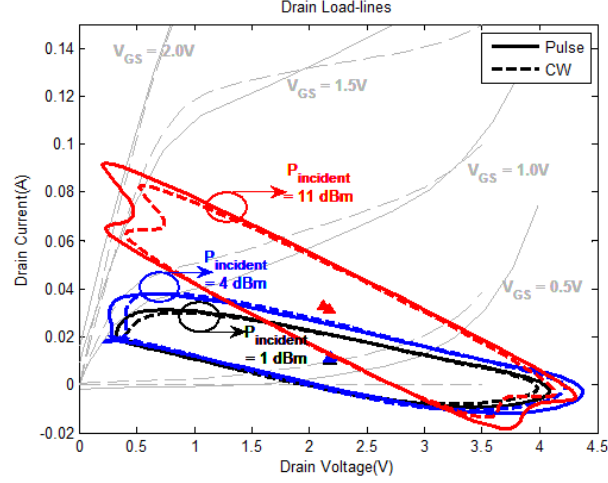


Figure 5.4: Drain load lines obtained for CW (dashed lines) and pulsed (plain lines) RF excitations with incident power of 1, 4, and 11 dBm and 1 % duty rate.

current of 11.8 mA (blue line with down triangles) which largely overestimates the self-heating in CW operation is seen to remain similar to the pulsed RF load line with the bias-point temperature (red line with up triangles). A resulting decrease in PAE of only 2% with this overestimated operating temperature is obtained as shown on Table 5.2. Clearly self-heating is not the dominating memory effect in this device. Now increasing the duty rate from 1% to 4% (magenta line with squares) is seen to produce a dynamic transfer characteristics and a dynamic load line which is bridging the gap between the ones obtained from CW and pulsed RF in Fig. 5.3 and 5.4. Also the increased duty rate yields a reduced PAE increase of 8% instead of 14% in Table 5.2.

In the particular case of the SOI technology considered the memory effects were seen to have a beneficial impact on the device performance. Namely under the transient peaks of the modulated RF excitations the transistor performed better than expected from operation at CW peak power while maintaining the same operating temperature. A similar

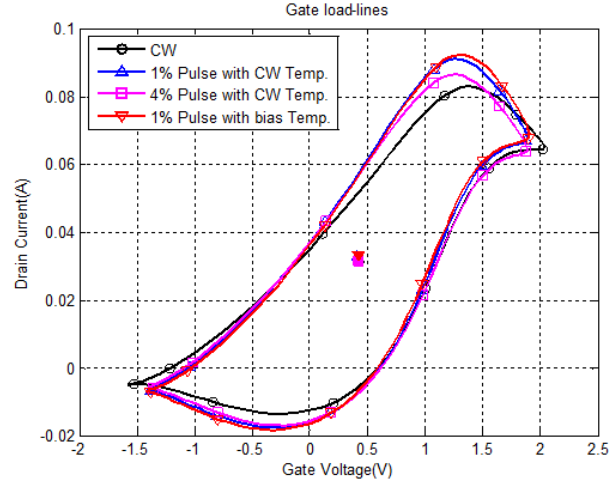


Figure 5.5: RF transfer obtained from 11 dBm CW (black line with circles) and pulsed RF excitations for 1% (blue and red lines with up and down triangles) and 4% (magenta line with square) duty rate and different hot-biases.

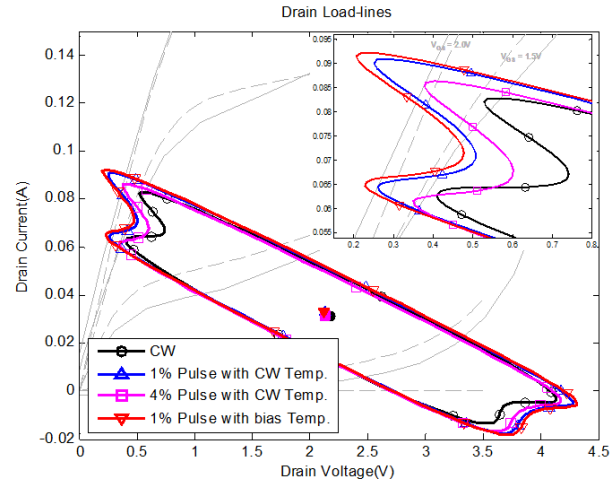


Figure 5.6: RF dynamic load line obtained for 11 dBm CW (black line with circles) and pulsed RF excitations for 1% (blue and red lines with low and upper triangles) and 4% (magenta line with square) duty rate and different hot biases.

Table 5.2: CW & Pulsed RF Transistor Performance @ 11 dBm

Duty Rate (%)	V_{GS} (V) (OFF/ON)	I_{DS} (mA) (OFF/ON)	P_{in} (mW)	P_{out} (mW)	Gain(dB)	PAE(%)
100	0.41	3/31.6	2.0	42.8	13.3	59.6
1	0.54/0.41	11.8/32.8	2.9	52.5	12.6	71.0
4	0.54/0.41	11.8/31.5	3.1	47.2	11.8	65.6
1	0.41/0.41	3/33.1	2.5	54.3	13.4	73.7

trend was observed on different devices and on different wafers. The precise origin of this memory effect is not known but several assertions can be made. Self-heating associated with the power dissipation in the device was verified not to be the controlling factor. This is explained by the relative low thermal resistance of the substrate used. The different gate voltage swing, drain voltage swing and drain current swing observed in Fig. 5.3 and 5.4 indicate that the gate and drain charges under the channel are different under pulsed and CW operations. The lower effective knee voltage observed in pulsed operation compared to CW operation could arise from a reduced effective channel resistance. The more negative drain current observed in Fig. 5.3 and 5.4 under pulsed operation which are associated with the improved PAE (negative power dissipation) is also indicative of a more efficient recovery of the charge stored in the channel. Presumably under unrealistic CW operation at peak power, the steady-state impact ionization may lead to charge accumulation adversely modifying the channel characteristics of the transistors.

Note that memory effects do not always have a beneficial impact like in the SOI MOS-FET considered in this chapter. For example trapping in GaN HEMTs induced by the transient peaks of the envelope of complex modulated RF excitations usually degrades the

device performance as was discussed. In all the cases the impact of memory effects on the transistor performance for RF signals with high PAPR needs to be carefully assessed. This is particularly needed for a more realistic design of power amplifiers relying on some type of load modulation for improved power efficiency. In such a case pulsed-RF load-pull from the proper hot bias condition will yield a more realistically representation of the peak power response of transistors excited with modulated signals with high peak to average power ratio.

5.2 Conclusion

The use of CW and pulsed RF large-signal measurements was investigated to quantify the impact of thermal and electrical memory effects upon the large-signal RF performance of transistors. Pulsed-RF load-pull measurements from realistic average hot bias condition should indeed yield a more realistically representation of the peak power response of transistors excited with modulated signals with high peak to average power ratio than can be obtained by CW load-pull measurements. For the SOI-MOSFET transistor considered an improved efficiency of 14 % was observed under pulsed operation compared to CW operation yielding a peak PAE of 73.7 %. The results obtained further demonstrated that slow memory effects do not respond solely to the average bias or temperature but can also depend on the RF load lines. Similar effects have been observed in GaN where intermittent high drain voltages can charges traps and induce knee voltage walk-out degrading in that case the average performance of the transistor.

CHAPTER 6

NONLINEAR DEVICE MODELING BASED ON ARTIFICIAL NEURAL NETWORK (ANN).

Various semiconductor materials (Si, GaAs, GaN, InP, etc) and device structures (SOI-MOSFET, MESFET, LDMOS, HEMT, etc) have been introduced and rapidly deployed in order to meet the high power and high-efficiency requirements of modern high frequency wireless communication [26] [29]. Under such circumstances, conventional modeling approaches (e.g., physically based or analytical closed-form empirical models) cannot keep up with the fast pace of technology development [30]. Measurement-based models, which by-pass this technology dependence in the model equations, were first introduced with linear measured S-parameters data and later on with nonlinear device models like the Root model and with nonlinear behavioral models such as X-parameters [31]. In all these model the functional dependence is typically implemented using spline functions. These measurement-based modeling approaches facilitate the development of large signal models and thus accelerate the design cycle for power amplifiers [32] [33] [34]. However, the main limitation in the use of spline function is (1) their poor extrapolation outside of the range of the data available and (2) their possible oscillation between data-points. The recently introduced artificial neural network (ANN) method provides very smooth functions

and non-vanishing derivatives of infinite order while guaranteeing a capability to fit any non-linearities of devices [35] [36] [37].

However, the model development cycle is still restricted due to the large number of S-parameter or nonlinear measurements needed for the model extraction. Indeed the measurement time rapidly increases as a dense grid on the intrinsic voltage plane is required for a better extraction of the model. The advent and successful deployment of non-linear vector measurement systems has induced an interest in using large signal characterization instead of conventional small signal characterization for device modeling. The main advantage of using large signal measurements for device modeling is that the model parameters are extracted under real device operating conditions and that the number of needed measurements can be reduced dramatically. The first trial using large signal measurements was conducted for the extraction of the nonlinear charges of a FET [38]. In [34] and [38], the gate and drain currents and charges extracted from large signal measurements were compared to the ones obtained from small signal measurements using an ideal analytical model to show the validity of the large signal approach. In addition to the excellent agreement obtained with the small signal results, the needed number of measurements was found to be reduced from N^2 to N times while obtaining the same intrinsic voltage coverage. However, the time consuming tuner characterization and implementation required in passive load-pull or the large number of amplitude and phase sweep measurements in the active load-pull system remain still a key sticking point in the model development cycle.

In this chapter we present a measurement-based quasi-static non-linear model for an SOS-MOSFET whose intrinsic nonlinear charges and current sources are extracted and verified from large signal measurements performed using the real-time active load-pull technique (RTALP) [39] [12]. Because RTALP data typically covers a wide range of device

operating point, the model can be extracted from only a single large signal measurement with an LSNA.

By directly comparing the measured and modeled load-lines, the proposed methodology allows for both extremely fast model extraction and verification procedures. Also, by comparing the output power and PAE in the whole Smith Chart, the model ability guarantees the excellent interpolation as well as extrapolation capabilities.

In addition the FET model is extended to include the parasitic-bipolar junction transistor (P-BJT) of the SOS-MOSFET so as to account for the associated low frequency memory effects and yield a model with improved accuracy.

6.1 Supportive Theory

6.1.1 Artificial Neural Network for Displacement Current

Artificial neural network (ANN) will be used in this work for representing the voltage dependence of the charges and current sources associated with the gate and drain of the SOS-MOSFET. ANN model are trained for the desired output (e.g. drain current) from various inputs (e.g. gate, drain voltages, and time derivative of voltages) by minimizing an error of cost function [42] [43]. The detail explanation of general ANN algorithm and cost function is included in an Appendix A.

Let us consider the intrinsic gate current. Typically the intrinsic gate current including the conduction and displacement currents is

$$i_G(t) = I_G(v_{GS}(t), v_{DS}(t)) + i_{disp}(t) \quad (6.1)$$

where I_G is the DC gate current and i_{disp} the displacement current defined as:

$$i_{disp}(t) = \frac{dQ_G(v_{GS}(t), v_{DS}(t))}{dt} = C_{gg} \frac{dv_{GS}}{dt} + C_{gd} \frac{dv_{DS}}{dt}. \quad (6.2)$$

If the DC gate current I_G has been characterized separately the instantaneous displacement current can be obtained using $i_{disp}(t) = i_G(t) - I_G(v_{GS}(t), v_{DS}(t))$.

To integrate the displacement current i_{disp} and obtain the gate charge Q_G we need to minimize the following cost function for a three layer ANN:

$$E_d = (i_{disp} - i_{ANN})^2 = \left(i_{disp} - \left[\frac{dv_{GS}}{dt} \cdot \Delta b_{31} + \frac{dv_{DS}}{dt} \cdot \Delta b_{32} \right] \right)^2 \quad (6.3)$$

In (6.3), Δb_{31} and Δb_{32} stand for the first derivatives of the ANN output $b_3 = Q_G$ (3rd layer) with respect to the network input 1 (v_{GS}) and 2 (v_{DS}), respectively.

The general case where the charge is extracted simultaneously with the conduction current is addressed in the appendix.

6.1.2 Time Domain Reconstruction for Real-time Tuning

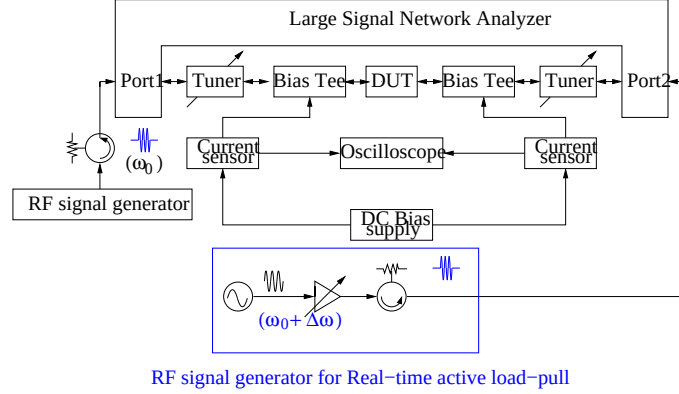


Figure 6.1: Real-time active load-pull system testbed.

The RTALP testbed used to characterize the device is shown in Fig. 6.1. The RF voltage and current waveforms at port i are given by

$$v_i(t) = a_{i,total}(t) + b_{i,total}(t) \quad (6.4)$$

$$i_i(t) = [a_{i,total}(t) - b_{i,total}(t)]/50 \quad (6.5)$$

where the time domain incident $a_{i,total}(t)$ and reflected $b_{i,total}(t)$ waveforms can be reconstructed from the measured frequency domain complex waves $a_i(n\omega_0 + p\Delta\omega)$ and $b_i(n\omega_0 + p\Delta\omega)$ measured by the LSNA using:

$$a_{i,total}(t) = \sum_{n=1}^N \sum_{p=-SSB}^{SSB} \text{Re} [a_i(n\omega_0 + p\Delta\omega) e^{j(n\omega_0 + p\Delta\omega)t}] \quad (6.6)$$

$$b_{i,total}(t) = \sum_{n=1}^N \sum_{p=-SSB}^{SSB} \text{Re} [b_i(n\omega_0 + p\Delta\omega) e^{j(n\omega_0 + p\Delta\omega)t}] \quad (6.7)$$

Also, the time-varying reflection coefficients at the n -th harmonic can be extracted using:

$$\Gamma_L(n\omega_0, t) = \frac{\sum_{p=-SSB}^{SSB} a_2(n\omega_0 + p\Delta\omega) e^{jp\Delta\omega t}}{\sum_{p=-SSB}^{SSB} b_2(n\omega_0 + p\Delta\omega) e^{jp\Delta\omega t}} \quad (6.8)$$

The fundamental RF output power at the output is also calculated by

$$P_{out}(\omega_0, t) = -\frac{1}{2} \sum_{p=-SSB}^{SSB} \sum_{q=-SSB}^{SSB} (v_2(\omega_0 + p\Delta\omega) \times i_2^*(\omega_0 + q\Delta\omega) e^{j(p-q)\Delta\omega t}) \quad (6.9)$$

6.2 Extrinsic parasitics extraction

Fig. 6.2 shows the device structure including the intrinsic transistor. To get access to the reference plane of the intrinsic transistor, all the parasitics shown in Fig. 6.2 were extracted in two separate steps as described below. Once all the parasitics are extracted, the intrinsic transistor model can be developed from the de-embedded large-signal measurement data. In the first step, the parasitics in the outer most layer $C_{1,2,3}$, $L_{1,2,3}$, and $R_{1,2}$ are extracted by

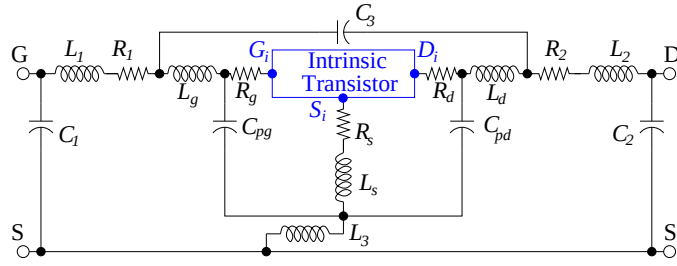


Figure 6.2: The circuit schematic of a typical large signal model of common source FET.

measuring the S-parameters of dummy structures which were fabricated on the same wafer as the tested devices. Using the extracted parasitics obtained in the first step, the remaining device parasitics ($C_{pg,pd}$, $L_{g,s,d}$, and $R_{g,s,d}$) are fitted using nonlinear least square method starting from physical estimated values.

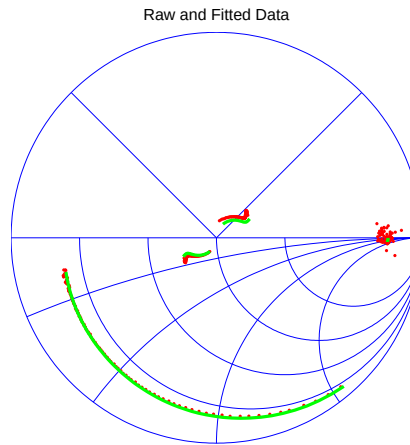


Figure 6.3: Measured(red dots) and fitted(green lines) S-parameters for one conduction dc-bias point.

Table 6.1: Extracted parasitics from dummny structure and using Cold-FET approach.

C_1 (fF)	19.0	C_2 (fF)	18.7
C_3 (fF)	3.22	C_{pg} (fF)	66.7
C_{pd} (fF)	175.0	R_1, R_g (Ω)	0.06, 6.5
R_2, R_d (Ω)	0.32, 1.10	R_3, R_s (Ω)	0, 0
L_1, L_g (pH)	61.1, 30.0	L_2, L_d (pH)	50.5, 1.35
L_3, L_s (pH)	2.94, 13.1		

The method used for this second extrinsic elements extraction is the Cold-FET [44] [45]. The idea of the Cold -FET method is simple: the device is biased with $V_{DS} = 0$ V. In this condition, there is no electric field in the channel. Then the gate voltage V_{GS} can be biased for values $> V_T$ or $< V_T$ so that the small-signal equivalent circuit can be simplified sufficiently to enable one to unambiguously identify the various elements. In our particular case we use the rigorous circuit model provided by the FET wave-equation theory [20] for $V_{GS} > V_T$. Fig. 6.3 shows the fitted and measured S-parameter results. The extracted data are summarized in Table I.

6.3 Modeling the intrinsic transistor

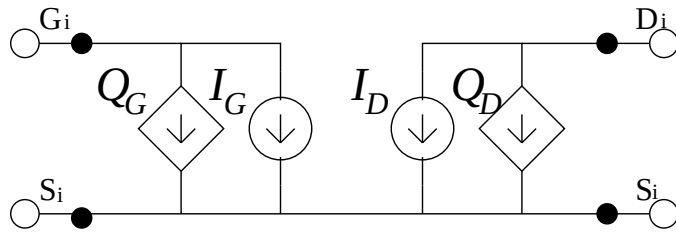


Figure 6.4: The quasi-static large-signal intrinsic model with no memory effects.

The voltage-controlled gate charge and current sources, and the voltage-controlled drain charge and current sources are shown in Fig. 6.4 for the intrinsic transistor. These charge and current sources are controlled by the instantaneous voltages of v_{GS} and v_{DS} . Therefore, the total currents in gate and drain sides are

$$i_G(t) = I_G(v_{GS}(t), v_{DS}(t)) + \frac{d}{dt}Q_G(v_{GS}(t), v_{DS}(t)) \quad (6.10)$$

$$i_D(t) = I_D(v_{GS}(t), v_{DS}(t)) + \frac{d}{dt}Q_D(v_{GS}(t), v_{DS}(t)) \quad (6.11)$$

Typically, the conduction current on the gate side is negligible and ignored due to insulating oxide between gate and channel in FETs and reverse-biased junction in HEMTs. Therefore, only Q_G , Q_D , and I_D will be investigated for the intrinsic transistor and then Eqn. 6.10 reduces to:

$$i_G(t) = \frac{d}{dt}Q_G(v_{GS}(t), v_{DS}(t)) \quad (6.12)$$

6.3.1 Quasi Static Model extraction

To acquire the drain conduction current source (I_D), DC-IV measurements were performed using a Keithley 4200 in the range of 0 V to 2 V with 0.1 V step for V_{gs} and 0 V to 3.5 V with 0.1 V step for V_{ds} . The 3-layers ANN structure is selected for all state functions and consists of two inputs, 20 non-linear neurons in the hidden layer, and one linear neuron at the output.

Fig. 6.5, 6.6, and 6.7 show the drain current, g_m , and g_{ds} comparison between the measurement (lines) and ANN model (circles). As shown in figures, the drain current as well as the first derivatives with respect to each voltage (g_m and g_{ds}) from the model are fitted very well with the measured data. This is achieved by using the same weight for the IV and its first derivatives in the ANN error cost function for the accurate prediction of harmonics and intermodulation [42].

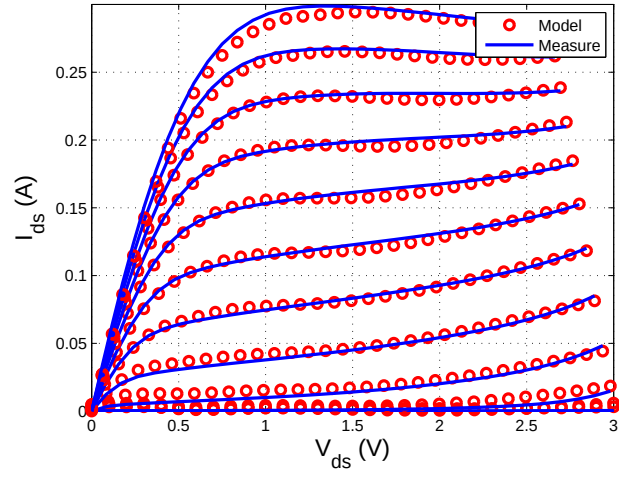


Figure 6.5: The I_D result comparison between measured (lines) and ANN model (circles).

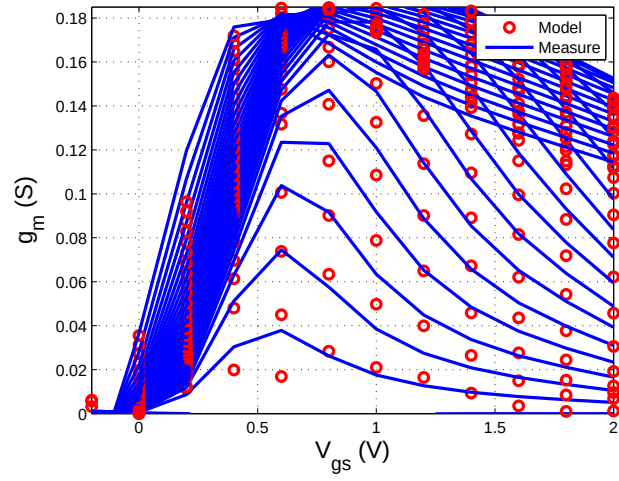


Figure 6.6: The g_m result comparison between measured (lines) and ANN model (circles).

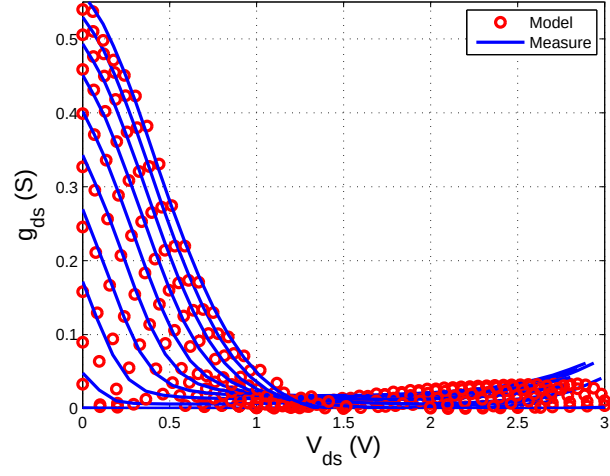


Figure 6.7: The g_{ds} result comparison between measured (lines) and ANN model (circles).

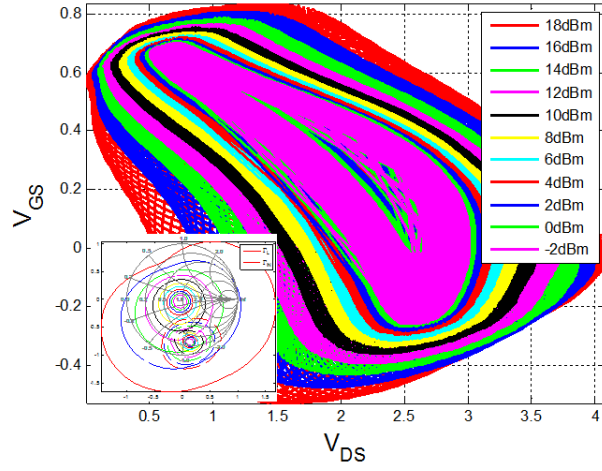


Figure 6.8: Dynamic load-lines of $v_{GS}(t)$ versus $v_{DS}(t)$ under different swept powers on the drain side. The inset shows the corresponding loci of Γ_L (solid line) and Γ_{IN} (dashed line).

The next step is to develop the ANN model for the gate and drain charges. The dynamic load-lines measured by the LSNA in v_{GS} - v_{DS} space are given in Fig. 6.8. As the swept power on the drain side increases, a wider voltage coverage is obtained due to larger amplitudes of load reflection coefficients. One of the advantages of the large-signal approach is that its measurement region can extend far beyond the conventional DC operation range of the device, and that extreme regions (e.g. breakdown) can be characterized without any device degradation. This holds because the frequency of the excited RF signal is typically above 1 GHz and the instantaneous voltage spends a relatively short time (nano scale) in the high voltage regions. However, the most important point is that the data used for the model extraction shown in Fig. 6.8 is measured from only a single LSNA measurement thanks to the real-time tuning. In previous large-signal approaches, such as passive load-pull or active load-pull systems, many different measurements from various load impedances, and RF power, and dc-bias conditions are necessary to obtain the same coverage. In this paper, 12 dBm (magenta) data is used for the model extraction in order to avoid the potentially unstable region when $|\Gamma_{IN}| > 1$.

Fig. 6.9 and 6.10 compares the LSNA and model simulation results for the gate and drain load-lines obtained at six different complex load impedances. The results shown provide a visual evaluation of the extraction accuracy. Note however that in this approach conduction and displacement current sources are extracted separately. First the ANN for the DC IV was extracted and next the ANN for the drain charge was extracted from the displacement current. If the ANN error cost functions for the conduction and displacement parts are combined together and the ANN networks simultaneously extracted, an improved fitting can be obtained. Table 6.2 compares the RMS values calculated from 100 different drain load-lines when the IV and charge ANNs are extracted separately and simultaneously.

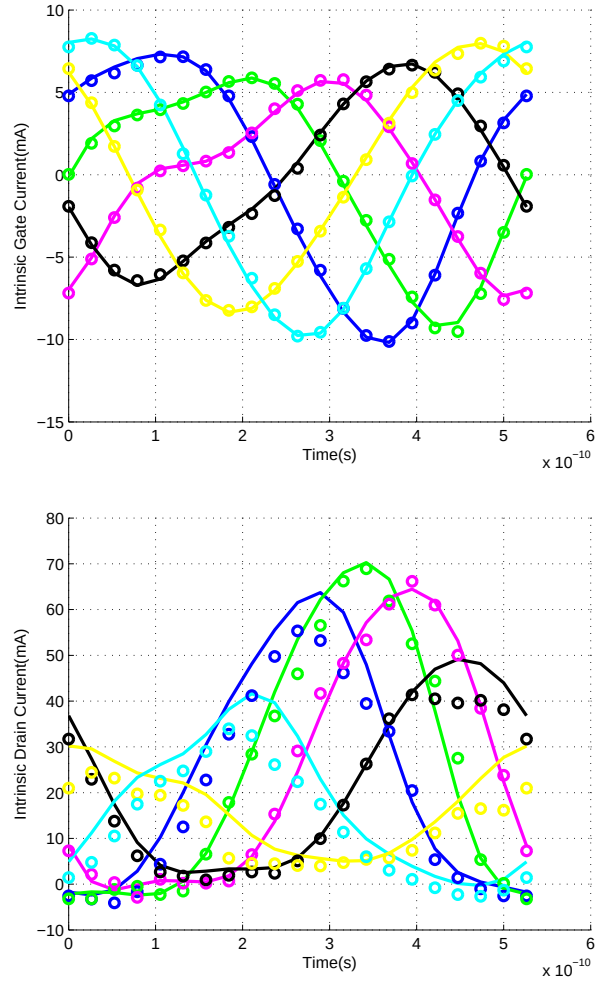


Figure 6.9: Measured(lines) and simulated (dots) gate(top) and drain(bottom) current waveforms from the extracted model separately.

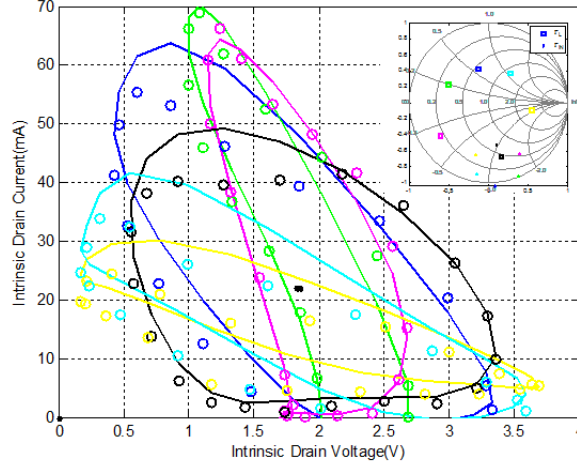


Figure 6.10: Dynamic load-lines at six different load impedances between the measurement (lines) and the model (circles) extracted separately. The inset shows the corresponding Γ_L (squares) and Γ_{IN} (crosses).

Note that different drain load-lines are used in the evaluation of the RMS for Prediction Case 1 (10 dBm) and Case 2 (14 dBm).

As is shown in Table 6.2, the RMS in the verification case is decreased by factor 2 as well as exhibits a large improvement in predictions for both Case 1 and 2. The improvement is due to the fact that for the DC IV characteristics used for the separate ANN extraction does not account for the memory effect associated with the P-BJT. This improvement suggest further modification of the model which are pursued in the next section.

6.3.2 Quasi static model extraction including memory effect

In the previous section, a quasi-static model with no explicit memory effects was developed and compared between different extraction approaches. Even though SOS-MOSFET devices exhibit less thermal effects than GaN HEMTs and negligible trap effects, they feature a different memory effect mechanism associated with their parasitic bipolar junction

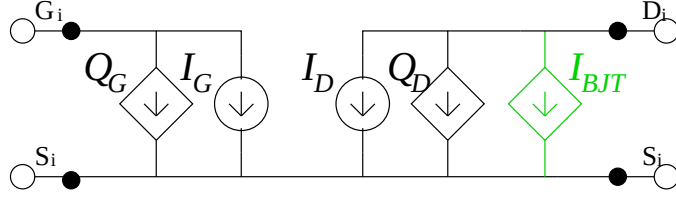


Figure 6.11: The quasi-static large-signal intrinsic model with parasitics BJT.

transistors (P-BJTs). This parasitic bipolar transistor is activated by the hole current generated generated by impact ionization at high drain voltages. However the P-BJT is a large device which does not responds at GHz frequencies and is thus only activated by the slower varying baseband component of the drain current and drain voltage. To account for and include this unwanted P-BJT into the model, ANN error cost function is further modified as follow:

$$i_D(v_{GS}, v_{DS}, V_{GS}, V_{DS}) = I_{D,RF}(v_{GS}, v_{DS}) + \frac{d}{dt}Q_D(v_{GS}, v_{DS}) + I_{BJT}(I_{D,RF}, V_{DS}) \quad (6.13)$$

Note that since the baseband current $I_{D,RF}$ is not available from the measurement, the baseband current $I_D(t)$ measured during RTALP is used instead. This is not a limitation since once the model extraction is completed the mapping between $I_{D,RF}$ and I_D is available.

Two approaches are used to model the P-BJT current. In the first method, P-BJT is assumed to be a linear function of the slowly varying baseband drain current and voltage induced by real-time tuning.

$$I_{BJT}(t) = I_{BJT,0} \quad (6.14)$$

$$+ \beta_{m,BJT} I_{D,RF}(t) + G_{d,BJT} [V_{DS}(t) - V_{DS}(DC)] \quad (6.15)$$

This linear expansion is well justify by the fact that the bias point varies little during RTALP as is demonstrated by its locus in Fig. (black dots around 1.7 V and 22 mA). A single layer ANN structure with a linear function is thus implemented and then added to the error cost function.

In the second approach the full non-linear characteristics of the P-JBT is represented by a separate ANN with $c_3 = I_{BJT}$ and extracted by simply adding the measured MOSFET DC IV to the RF load line data used for the extraction. Indeed for a DC (or low frequency < 100 MHz) excitation the model satisfies:

$$I_D(V_{GS}, V_{DS}) = I_{D,RF}(V_{GS}, V_{DS}) + I_{BJT}(I_{D,RF}, V_{DS}) \quad (6.16)$$

The update of cost function to include the contribution is $c_3 = I_{BJT}$ is discussed in appendix. The calculation of the additional Jacobian is similar to that of the drain current.

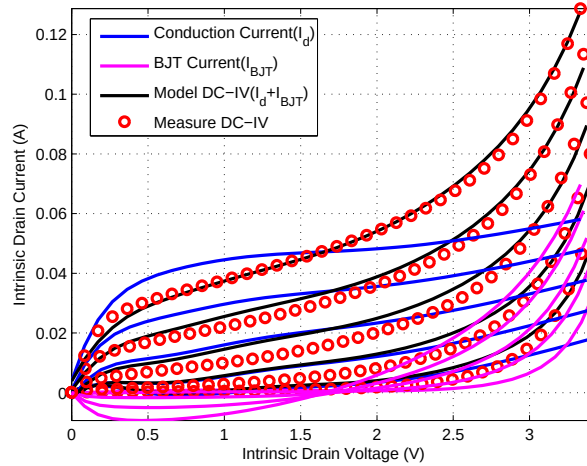


Figure 6.12: The measured DC-IV (circles) and extracted drain conduction currents (lines) with non-linear P-BJT currents.

Table 6.2: RMS Comparison

	Verification	Prediction Case 1	Prediction Case 2
Separate extraction of FET charge and DC IV	6.55	5.25	9.18
Simultaneous extraction of FET charge and RF IV	3.08	3.69	5.65
Simultaneous extraction of FET and linearized BJT	2.77	3.65	5.44
Simultaneous extraction of FET and non-linear BJT IV	3.53	3.48	6.03

Fig. 6.12 shows measured and model simulated DC-IV including conduction and non-linear P-BJT currents. In the high drain voltage region, impact ionization brings about a substantial drain current increase due to the high electric field in the channel. The extracted P-BJT reflects well this exponential behavior so that the combined DC-IV from the model exhibit a reasonably good agreement with the measured DC-IV.

Table 6.2 shows a comparison in the RMS obtained over 100 different drain load-lines between linear and non-linear P-BJT models as well as different extraction models in the previous section. As removing the measurement error, the RMS exhibits a noticeable improvement in verification and both prediction cases for the linearized BJT model. A mixed result of small improvement and small degradation is observed in both verification and prediction using the nonlinear BJT model. The degradation in prediction for Case 2 may be due to the fact that the BJT model slightly exaggerates the impact of the baseband currents. Note that in Case the model is asked to predict the device performance at larger output power than was used for the extraction.

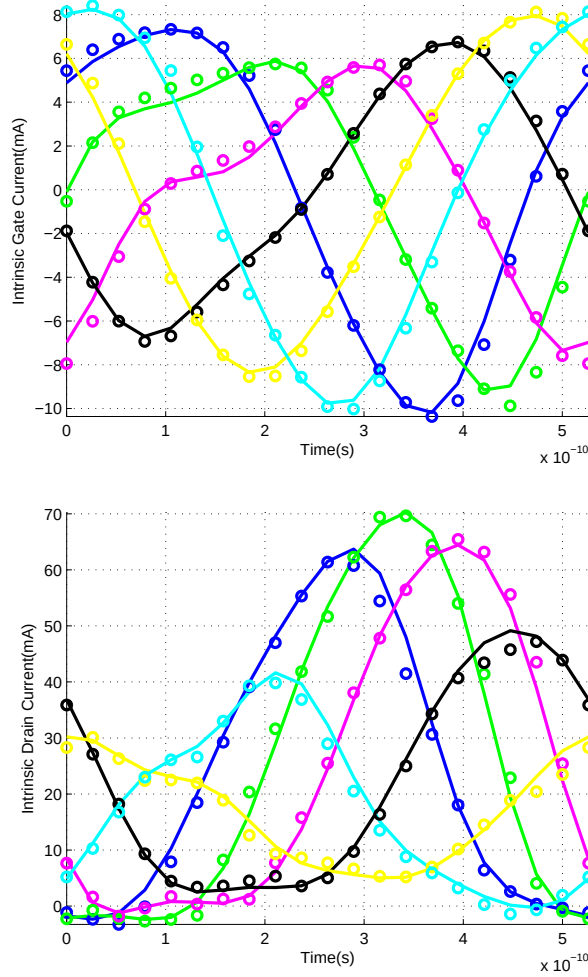


Figure 6.13: Measured(lines) and simulated (dots) gate(top) and drain(bottom) current waveforms from the simultaneously extracted model including linear P-BJT.

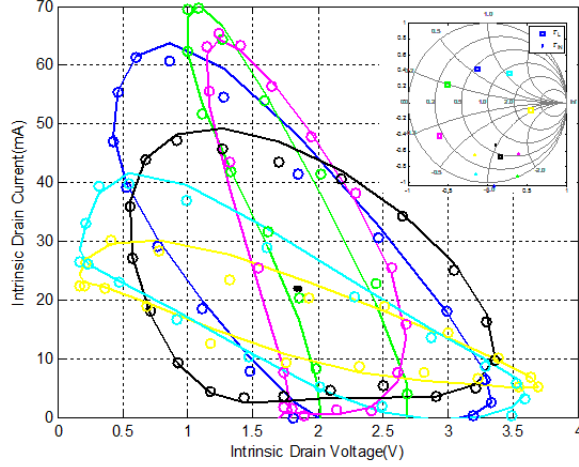


Figure 6.14: Dynamic load-lines at six different load impedances between the measurement (lines) and the model extracted simultaneously with linear P-BJT implementation (circles). The inset shows the corresponding Γ_L (squares) and Γ_{IN} (crosses).

Fig. 6.13 and 6.14 shows LSNA and model simulation with linear P-BJT implementation gate and drain currents waveforms for the same six different complex load impedances used in Fig. 6.10. By comparing between Fig. 6.10 and 6.14, a noticeable improvements is observed in all six different load-lines as predicted by the RMS figure of merit.

Q_G and Q_D are also extracted for the same measurement and presented in Fig. 6.15. As mentioned before, a graceful prediction of the gate and drain charges outside the data extraction region is observed thanks to the good ANN extrapolation capability.

To further test the capability of this model developed from a single LSNA measurement, Fig. 6.16, 6.17, and 6.18 present a comparison of the between modeled and measured results for the input power(P_{in}), output power(P_{out}), DC dissipated power(P_{dc}), and power added efficiency(PAE) at 20 different complex load impedances. Note that the 20 different points were selected from a single RTLAP measurement and there RTLAP measurement

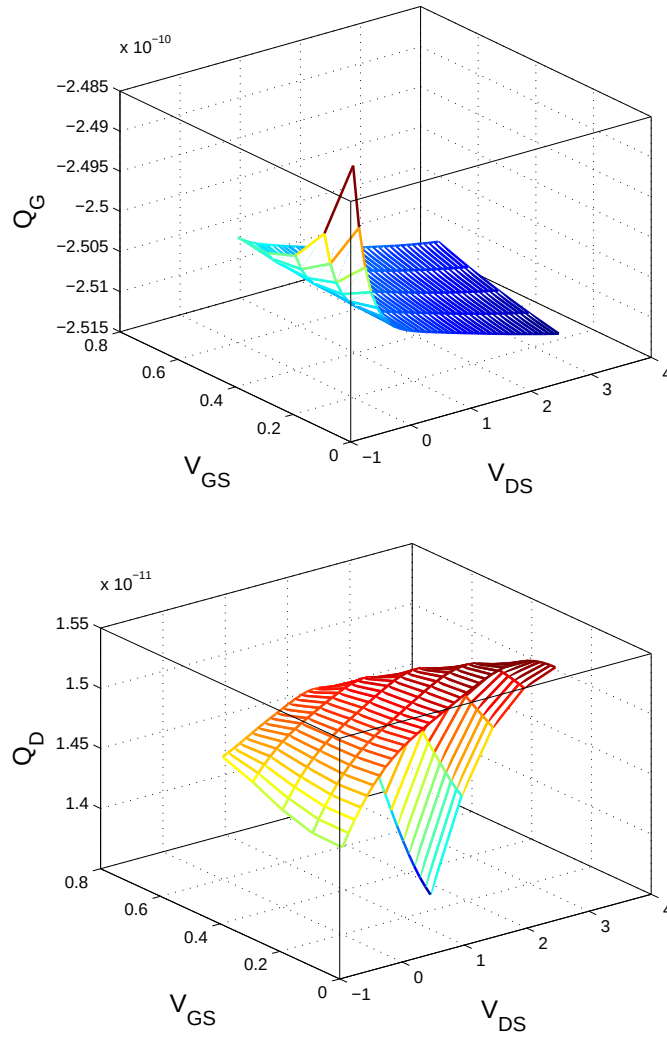


Figure 6.15: Extracted Q_G (top) and Q_D (bottom) from sigle dynamic large-signal measurement.

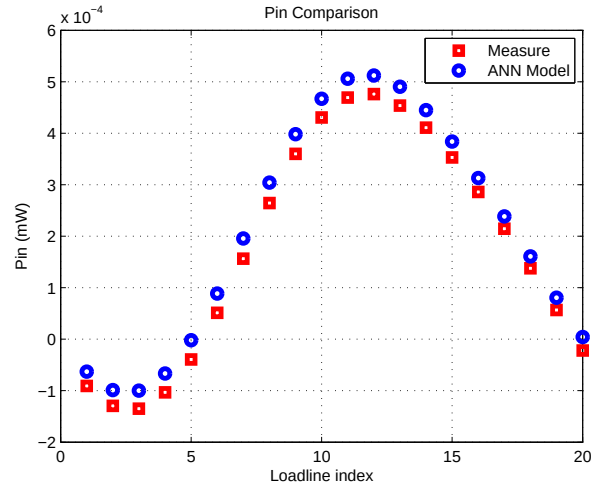


Figure 6.16: Measured (red squares) and modeled (blue circles) input power comparison at 20 different complex load impedances.

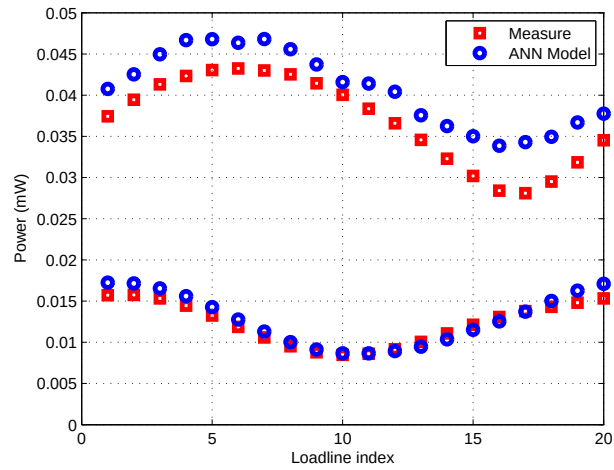


Figure 6.17: Measured (red squares) and modeled (blue circles) P_{dc} and P_{out} comparison at 20 different complex load impedances.

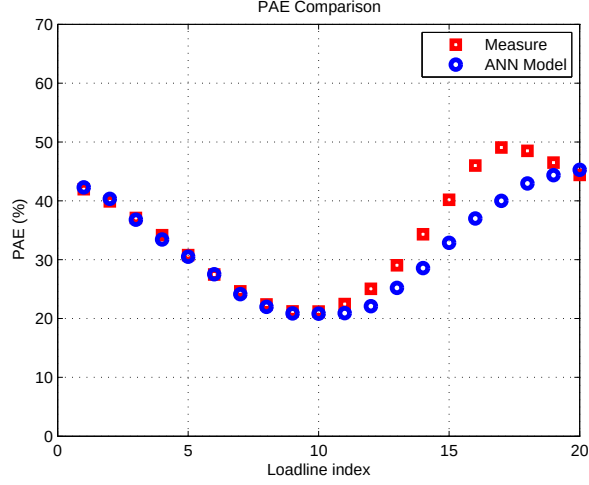


Figure 6.18: Measured (red squares) and modeled (blue circles) PAE comparison at 20 different complex load impedances.

data were not used in the ANN training. As shown in these figures, all the modeled results and most of the measured results are approximately following a sinusoidal shape as the phase of the reflection coefficient $\Gamma_L(\omega)$ is swept by the real-time tuning along its closed contour.

Finally, a comparison of the model large-signal prediction with respect to measured output power and PAE on the Smith Chart is presented in Fig. 6.19 and 6.20. The model semi-quantitatively agrees with the measured output power and PAE on the extracted whole Smith Chart. The optimum load impedance points for output power and PAE located at different load points under this modulated RF measurement are reasonably well predicted. This indicates that the proposed model reflect well the device behavior at both RF and low frequencies.

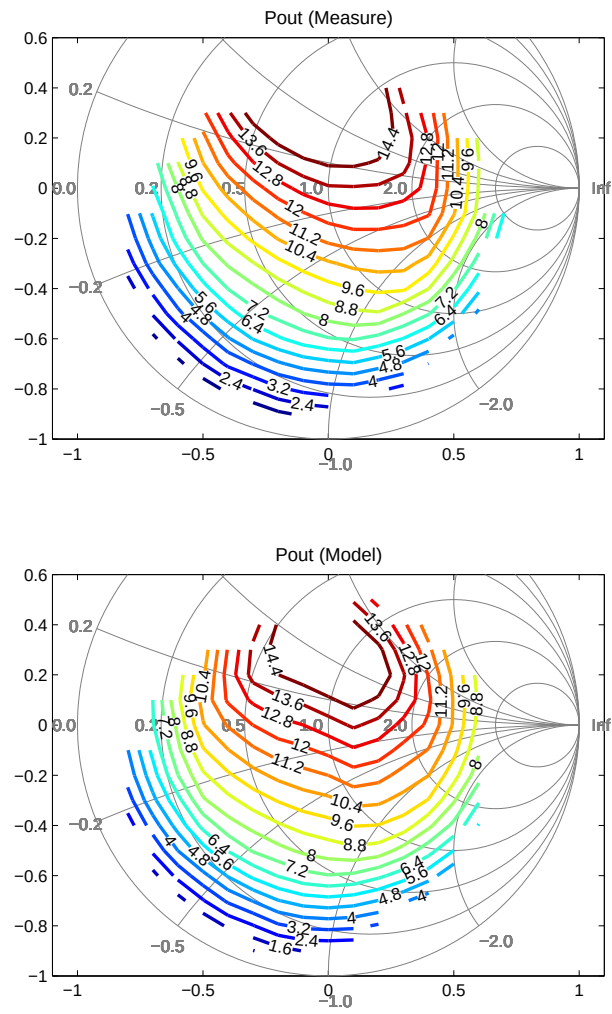


Figure 6.19: Measured(top) and simulated(bottom) output power contours.

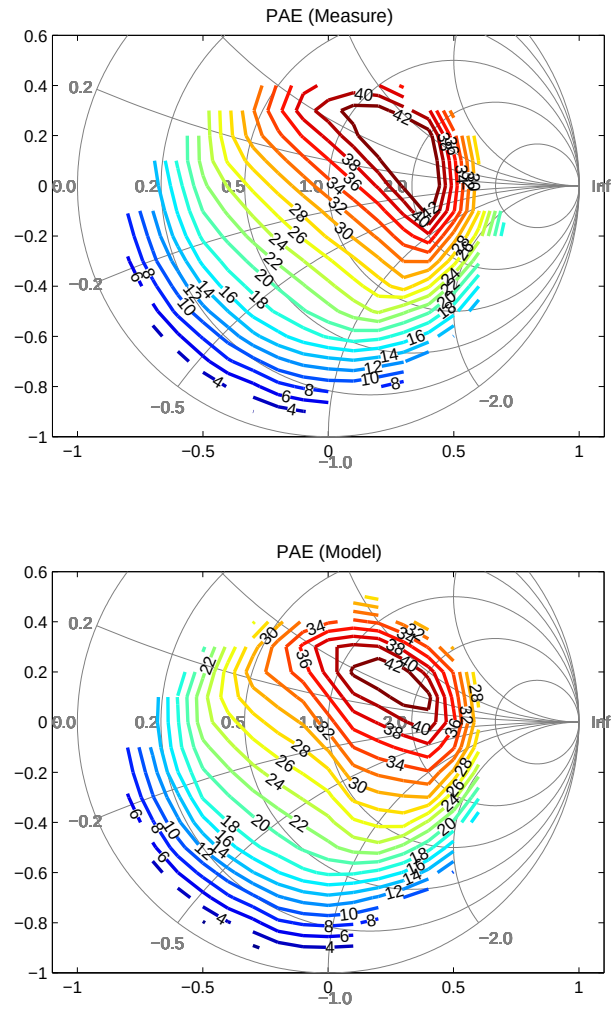


Figure 6.20: Measured(top) and simulated(bottom) PAE contours.

6.4 Conclusion

A measurement based quasi-static non-linear model with ANN approach has been presented. A fast and easy model development process based on a single large-signal measurement has been demonstrated thanks to the unique characteristics of real-time tuning. A new ANN error cost function for conduction current, charge sources, and parasitic bipolar transistor current has also been introduced and then implemented for the ANN model extraction. Load-line comparison at different complex load impedances shows the validity of the proposed methodology. Furthermore, the similar sinusoidal trend obtained in the comparison of the variation of the PAE, P_{dc} , and P_{out} with the RTALP phase sweeping for the measured and modeled results demonstrates the ability of the model to incorporate low-frequency dispersions of devices into the RF model.

CHAPTER 7

CONCLUSION AND FUTURE WORK

The demand for accurate transistor model has been continuously increased as various transistor structures and materials for the power amplifier design are introduced to meet the stringent needs of the wireless PA designer community. However, the operation at high power and high frequencies of the RF and microwave power amplifier designed bring about lots of challenges. As the operating frequency increases, the transistor itself becomes large so that the electrical behavior should be considered in terms of distributed circuits. As the power requirement for PAs increases, the transistors' thermal effect and other dispersive effects (traps and bipolar transistors) need to be characterized with an appropriate measurement system and then modeled accurately to enable high power designs. In view of these challenges, the goal of this thesis is to develop RF measurement systems capable of characterizing the device memory effects involved in high power high frequency RF power amplifiers as well as demonstrate how a large-signal device model can be efficiently extracted from these characterization measurements.

To address the need of wide bandwidth communication, a broadband measurement testbed was developed with an LSNA. This testbed presented in Chapter 2 is intended for characterizing PAs using periodically modulated (multitone) signals often referred as multisines. In the experiment conducted a multisine with 13 tones distributed over 15 MHz

was used which provided a PAPR of 6 dB. This vector measurement enables one to characterize the nonlinear response of the PA in its 3rd and 5th order inter-modulation bands at the fundamental frequency. The broadband vector results at the second harmonic obtained in the same measurements provides additional invaluable information for the multi-harmonic optimization of the efficiency of the Doherty PA under test. It is believed that the multi-harmonic broadband vector characterization provided by the reported testbed should assist with the development and verification of multi-harmonic behavioral models for wideband modulated signals.

In Chapter 3, a pulsed-RF RTALP measurement system was also implemented with the LSNA. The recently reported multiple recording technique was used in the pulsed-RF RTALP measurement system for the acquisition of pulsed-RF signal without any loss of dynamic range. This testbed was then utilized for exploring the 1dB compression gain plane for the whole Smith chart. The new pulsed-RF RTALP method provides great saving in measurement time and effort compared to the conventional passive or active load pull methods. The experiments were conducted with a passivated AlGaIn/GaN HEMT transistor to determine the optimal load terminations for maximum gain and output power in pulsed excitation operation. It was further verified that the pulsed-RF RTALP yielded the same results as CW RTALP under low power RF operation where the AlGaIn/GaN HEMT exhibit negligible memory effects. Also comparison the results obtained at three critical load points, with those obtained from CW and pulsed constant phase ALP measurements, it was verified that the memory effects caused by the real-time tuning are significantly reduced. This new methodology provides great potential for speeding up the characterization of nonlinear power amplifier design while removing memory effects including self-heating and trap.

In Chapter 4 and 5, various memory effects were characterized by using pulse measurement system. First, it was demonstrated that the short capture time of the traps at high voltages can be considerably lengthened through the use of μs large-signal pulsed-RF excitations with duty rate as low as 0.001%. In addition, by comparing the device performance for two different load impedances, it was shown that (1) the average power dissipation determines the internal device temperature while (2) the instantaneous high drain voltage is responsible for the enhanced trapping. Moreover, the use of CW and pulsed RF large-signal measurements was used to investigate the impact of thermal and electrical memory effects upon the large-signal RF performance of transistors. Pulsed-RF load-pull measurements from realistic average hot bias condition should indeed yield a more realistically representation of the peak power response of transistors excited with modulated signals with high peak to average power ratio than can be obtained by CW load-pull measurements. For an SOI-MOSFET transistor considered an improved efficiency of 14 % was observed under pulsed operation compared to CW operation yielding a peak PAE of 73.7 %. The results obtained further demonstrated that slow memory effects do not respond solely to the average bias or temperature but can also depend on the RF load lines. Similar effects have been observed in GaN where intermittent high drain voltages can charge traps and induce knee voltage walk-out degrading in that case the average performance of the transistor.

Given the developed measurement system and the experimental data available, a measurement based quasi-static non-linear model based on an artificial neural network (ANN) was presented in Chapter 6. A fast and easy model development process based on a single large-signal measurements has been demonstrated thanks to the real-time tuning measurement system. A new ANN cost function for the simultaneous extraction of the MOSFET current-voltage, MOSFET charge-voltage and parasitic BJT characteristics was introduced

and then implemented for the ANN model extraction. Load-line comparisons at different complex load impedances demonstrated the validity of the proposed methodology. Moreover, the same trend in PAE, P_{dc} , and P_{out} comparison observed in the real-time tuning (phase sweep) between the measured and modeled data demonstrated the capability of the model in incorporating the low-frequency dispersions of the device into the model.

Future work will make use of the new measurement techniques to incorporate additional memory effects in the transistors models. Measurements at multiple bias points could be pursued to fully characterize the memory effects. In GaN device the trap dynamics and the observed cyclo-stationary effects could be incorporated in the model. In conclusion a wide variety of research investigations in the characterization and nonlinear modeling of RF power transistors in the presence of low-frequency memory effect remain thus possible.

APPENDIX A

ERROR COST FUNCTION DERIVATION FOR ARTIFICIAL NEURON NETWORK (ANN)

Artificial neural network(ANN) have been rapidly developed and known to be a useful tool for various simulation fields such as electromagnetic(EM), signal integrity, transistor electrical parameter, and more [43]. Especially, if an analytical model, which is typically computationally expensive, is not available, the ANN approach provides an alternative solution with less computational price.

The most general cost function used in this work is of the form:

$$E = \frac{1}{P} \sum_{i=1}^P e^2(i)$$
$$\text{with } e(i) = i_D(i) - \{a_3(i)$$
$$+ \left[\frac{dv_{GS}(i)}{dt} \cdot \Delta b_{31} + \frac{dv_{DS}(i)}{dt} \cdot \Delta b_{32} \right]$$
$$+ c_3(i)\}$$

where i is used to indicate the time sampling at time t_i .

E is the error cost functions for the sum of drain conduction, displacement and P-BJT currents. Also, $i_D(i)$ is the LSNA measured drain current, and $a_3 = I_D$, $b_3 = Q_D$, and $c_3 = I_{BJT}$ are the desired ANN outputs from the conduction current, FET charge, and P-BJT conduction current, respectively.

Table A.1: ANN Matrix Notation and Size		
Description	Symbol	Matrix size
Input	$\bar{a}_1$	$(P \times N+1)$
First layer weight	$\bar{w}_1$	$(M \times N+1)$
$\bar{a}_1 \bar{w}_1^t$	$\overline{net}_1$	$(P \times M)$
$f_1(\overline{net}_1)$	$\bar{a}_2$	$(P \times M)$
Second layer weight	$\bar{w}_2$	$(1 \times M)$
$\bar{a}_2 \bar{w}_2^t$	$\overline{net}_2$	$(P \times 1)$
Output or $f_2(\overline{net}_2)$	$\bar{a}_3$	$(P \times 1)$

The P-BJT current will have the same Jacobian derivation notation with the drain conduction current. Therefore, only the drain current derivation will be investigated.

A.1 Backpropagation Algorithm for the Conduction Current

As shown in Fig. A.1, the ANN structure consists of three different layers: the input, hidden and output layers which are interconnected by the two different weights $\bar{w}_1$ and $\bar{w}_2$. The two different weights, therefore, need to be updated during iteration by comparing the past and present value of the output (drain current) and its derivatives. This is done with the help of the Jacobian which involves the derivative of the ANN nodes with respect to the weights $\bar{w}_1$ and $\bar{w}_2$ respectively.

In Table A.1, P and M are the number of data for the input and of hidden layers respectively.

A.1.1 At the output layer

At the output layer, the derivative of network output($\bar{a}_3$) with respect to $\bar{w}_2$ is

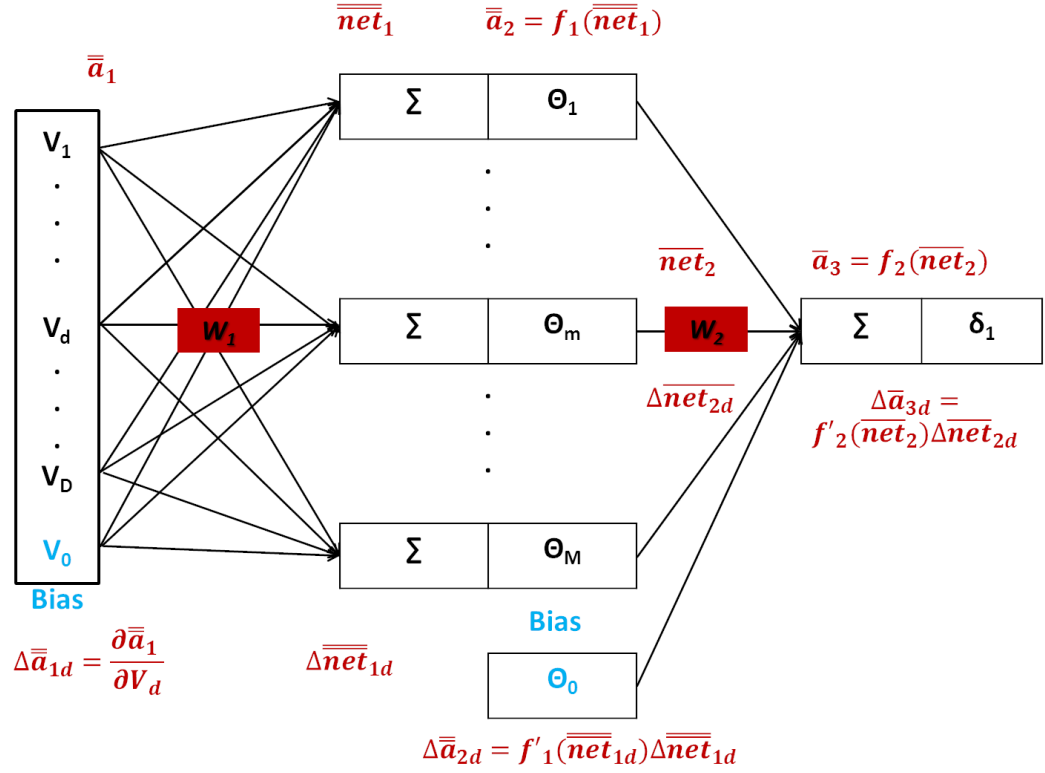


Figure A.1: ANN Diagram

$$\begin{aligned}
 J_2(j) &= \frac{\partial E_0}{\partial w_2(j)} = -\frac{2}{P} \sum_{i=1}^P e_0(i) \frac{\partial a_3(i)}{\partial w_2(j)} \\
 &= -\frac{2}{P} \sum_{i=1}^P e_0(i) \frac{\partial}{\partial w_2(j)} f_2[net_2(i)] \\
 &= -\frac{2}{P} \sum_{i=1}^P e_0(i) \frac{\partial}{\partial w_2(j)} f_2 \left[\sum_{m=1}^M a_2(i, m) w_2(m) \right] \\
 &= -\frac{2}{P} \sum_{i=1}^P e_0(i) f'_2[net_2(i)] a_2(i, j) \\
 &= -\frac{2}{P} \sum_{i=1}^P \delta_2(i) a_2(i, j)
 \end{aligned}$$

where $\delta_2(i) = e_0(i) f'_2[net_2(i)]$

Therefore,

$$\bar{J}_2 = \frac{\partial E_0}{\partial \bar{w}_2} = -\frac{2}{P} \bar{\delta}_2^T \bar{a}_2 \quad (\text{A.1})$$

with $\bar{\delta}_2 = \bar{e}_0 * f'_2 [\bar{\text{net}}_2]$

A.1.2 At the hidden layer

In the hidden layer, error cost function will be the same except that the derivative of output is with respect to $\bar{w}_1$ instead of $\bar{w}_2$ at the output layer.

Therefore, the equation is:

$$\begin{aligned} J_1(j, k) &= \frac{\partial E_0}{\partial w_1(j, k)} = -\frac{2}{P} \sum_{i=1}^P e_0(i) \frac{\partial a_3(i)}{\partial w_1(j, k)} \\ &= -\frac{2}{P} \sum_{i=1}^P e_0(i) \frac{\partial}{\partial w_1(j, k)} f_2[\text{net}_2(i)] \\ &= -\frac{2}{P} \sum_{i=1}^P e_0(i) \frac{\partial}{\partial w_1(j, k)} f_2 \left[\sum_{m=1}^M a_2(i, m) w_2(m) \right] \\ &= -\frac{2}{P} \sum_{i=1}^P e_0(i) \frac{\partial}{\partial w_1(j, k)} f_2 \left[\sum_{m=1}^M f_1[\text{net}_1(i, m)] w_2(m) \right] \\ &= -\frac{2}{P} \sum_{i=1}^P e_0(i) \frac{\partial}{\partial w_1(j, k)} f_2 \left[\sum_{m=1}^M f_1 \left[\sum_{n=1}^{N+1} a_1(i, n) w_1(m, n) \right] w_2(m) \right] \\ &= -\frac{2}{P} \sum_{i=1}^P e_0(i) f'_2[\text{net}_2(i)] w_2(j) f'_1[\text{net}_1(i, j)] a_1(i, k) \\ &= -\frac{2}{P} \sum_{i=1}^P \delta_2(i) w_2(j) f'_1[\text{net}_1(i, j)] a_1(i, k) \\ &= -\frac{2}{P} \sum_{i=1}^P \delta_1(i, j) a_1(i, k) \end{aligned}$$

where $\delta_1(i, j) = \delta_2(i) w_2(j) f'_1[\text{net}_1(i, j)]$

Therefore,

$$\begin{aligned}\bar{\bar{J}}_1 &= \frac{\partial E_0}{\partial \bar{w}_1} = -\frac{2\bar{\bar{\delta}}_1^T}{p} \bar{\bar{a}}_1 \\ \text{with } \bar{\bar{\delta}}_1 &= \bar{\delta}_2 \bar{w}_2 \cdot * f'_1 \left[\bar{\bar{\text{net}}}_1 \right]\end{aligned}\tag{A.2}$$

A.2 Backpropagation Algorithm for first derivative of Drain current

In previous section, the error cost function for drain conduction current was investigated at two different layers. For the displacement current, the first derivatives with respect to inputs(v_{GS} and v_{DS}) will be used in the error cost function.

A.2.1 At the output layer

The derivative of E_d with respect to $\bar{w}_2$ is

$$\begin{aligned}\Delta J_2(j) &= \frac{\partial E_d}{\partial w_2(j)} = -\frac{2}{P} \sum_{i=1}^P e_d(i) \frac{\partial \Delta a_{3d}(i)}{\partial w_2(j)} \\ &= -\frac{2}{P} \sum_{i=1}^P e_d(i) \frac{\partial}{\partial w_2(j)} \{f'_2[\text{net}_2(i)] \Delta \text{net}_{2d}(i)\} \\ &= -\frac{2}{P} \sum_{i=1}^P e_d(i) \frac{\partial}{\partial w_2(j)} \left\{ f'_2 \left[\sum_{m=1}^M a_2(i, m) w_2(m) \right] \sum_{m=1}^M \Delta a_{2d}(i, m) w_2(m) \right\} \\ &= -\frac{2}{P} \sum_{i=1}^P e_d(i) \{f'_2[\text{net}_2(i)] \Delta a_{2d}(i, j) + \Delta \text{net}_{2d}(i) f''_2[\text{net}_2(i)] a_2(i, j)\} \\ &= -\frac{2}{P} \sum_{i=1}^P \delta_{2d}^A(i) \Delta a_{2d}(i, j) + \delta_{2d}^B(i) a_2(i, j)\end{aligned}$$

where,

$$\delta_{2d}^A(i) = e_d(i) f'_2[\text{net}_2(i)]$$

$$\delta_{2d}^B(i) = e_d(i) \Delta \text{net}_{2d}(i) f''_2[\text{net}_2(i)]$$

Therefore we have:

$$\begin{aligned}\Delta \bar{J}_2 &= \frac{\partial E_d}{\partial \bar{w}_2} = -\frac{2}{P} \left\{ \left(\bar{\delta}_{2d}^A \right)^T \Delta \bar{a}_{2d} + \left(\bar{\delta}_{2d}^B \right)^T \bar{a}_2 \right\} \\ &\quad \text{with } \bar{\delta}_{2d}^A = \bar{e}_d \cdot * f_2' [\overline{\text{net}_2}] \\ &\quad \bar{\delta}_{2d}^B = \bar{e}_d \cdot * \Delta \overline{\text{net}_{2d}} \cdot * f_2'' [\overline{\text{net}_2}]\end{aligned}\tag{A.3}$$

A.2.2 At the hidden layer

For the hidden layer the derivative of E_d with respect to $\bar{w}_1$ is:

$$\begin{aligned}\Delta J_1(j, k) &= \frac{\partial E_d}{\partial w_1(j, k)} = -\frac{2}{P} \sum_{i=1}^P e_d(i) \frac{\partial \Delta a_{3d}(i)}{\partial w_1(j, k)} \\ &= -\frac{2}{P} \sum_{i=1}^P e_d(i) \frac{\partial}{\partial w_1(j, k)} \{ f_2' [\text{net}_2(i)] \Delta \text{net}_{2d}(i) \} \\ &= -\frac{2}{P} \sum_{i=1}^P e_d(i) \{ \Delta \text{net}_{2d}(i) \frac{\partial}{\partial w_1(j, k)} f_2' [\text{net}_2(i)] \\ &\quad + f_2' [\text{net}_2(i)] \frac{\partial}{\partial w_1(j, k)} \Delta \text{net}_{2d}(i) \}\end{aligned}$$

For the first term,

$$\begin{aligned}&\Delta \text{net}_{2d}(i) \frac{\partial}{\partial w_1(j, k)} f_2' [\text{net}_2(i)] \\ &= \Delta \text{net}_{2d}(i) \frac{\partial}{\partial w_1(j, k)} f_2' \left[\sum_{m=1}^M a_2(i, m) w_2(m) \right] \\ &= \Delta \text{net}_{2d}(i) \frac{\partial}{\partial w_1(j, k)} f_2' \left[\sum_{m=1}^M f_1 [\text{net}_1(i, m)] w_2(m) \right] \\ &= \Delta \text{net}_{2d}(i) \frac{\partial}{\partial w_1(j, k)} f_2' \left[\sum_{m=1}^M f_1 \left[\sum_{n=1}^{N+1} a_1(i, n) w_1(m, n) \right] w_2(m) \right] \\ &= \Delta \text{net}_{2d}(i) f_2'' [\text{net}_2(i)] w_2(j) f_1' [\text{net}_1(i, j)] a_1(i, k)\end{aligned}$$

For the second term,

$$\begin{aligned}
& f'_2 [net_2(i)] \frac{\partial}{\partial w_1(j, k)} \Delta net_{2d}(i) \\
&= f'_2 [net_2(i)] \frac{\partial}{\partial w_1(j, k)} \sum_{m=1}^M \Delta a_{2d}(i, m) w_2(m) \\
&= f'_2 [net_2(i)] \sum_{m=1}^M \frac{\partial}{\partial w_1(j, k)} \{f'_1 [net_1(i, m)] \Delta net_{1d}(i, m)\} w_2(m) \\
&= f'_2 [net_2(i)] \sum_{m=1}^M \{f'_1 [net_1(i, m)] \frac{\partial}{\partial w_1(j, k)} \Delta net_{1d}(i, m) \\
&\quad + \Delta net_{1d}(i, m) \frac{\partial}{\partial w_1(j, k)} f'_1 [net_1(i, m)]\} w_2(m) \\
&= f'_2 [net_2(i)] \left\{ \sum_{m=1}^M f'_1 [net_1(i, m)] w_2(m) \frac{\partial}{\partial w_1(j, k)} \left[\sum_{n=1}^{N+1} \Delta a_{1d}(i, n) w_1(m, n) \right] \right. \\
&\quad \left. + \sum_{m=1}^M \Delta net_{1d}(i, m) w_2(m) \frac{\partial}{\partial w_1(j, k)} f'_1 \left[\sum_{n=1}^{N+1} a_1(i, n) w_1(m, n) \right] \right\} \\
&= f'_2 [net_2(i)] w_2(j) \{f'_1 [net_1(i, j)] \Delta a_{1d}(i, k) + \Delta net_{1d}(i, j) f''_1 [net_1(i, j)] a_1(i, k)\}
\end{aligned}$$

Therefore,

$$\begin{aligned}
\Delta J_1(j, k) &= \frac{\partial E_d}{\partial w_1(j, k)} \\
&= -\frac{2}{P} \sum_{i=1}^P e_d(i) f'_2 [net_2(i)] w_2(j) f'_1 [net_1(i, j)] \Delta a_{1d}(i, k) \\
&\quad + e_d(i) \{f'_2 [net_2(i)] w_2(j) \Delta net_{1d}(i, j) f''_1 [net_1(i, j)] \\
&\quad + \Delta net_{2d}(i) f''_2 [net_2(i)] w_2(j) f'_1 [net_1(i, j)]\} a_1(i, k) \\
&= -\frac{2}{P} \sum_{i=1}^P \delta_{2d}^A(i) w_2(j) f'_1 [net_1(i, j)] \Delta a_{1d}(i, k) \\
&\quad + \delta_{2d}^A(i) w_2(j) \Delta net_{1d}(i, j) f''_1 [net_1(i, j)] a_1(i, k) + \delta_{2d}^B(i) w_2(j) f'_1 [net_1(i, j)] a_1(i, k) \\
&= -\frac{2}{P} \sum_{i=1}^P \delta_{1d}^A(i, j) \Delta a_{1d}(i, k) + \delta_{1d}^B(i, j) a_1(i, k)
\end{aligned}$$

where

$$\delta_{1d}^A(i, j) = \delta_{2d}^A(i)w_2(j)f_1' [net_1(i, j)]$$

$$\delta_{1d}^B(i, j) = \delta_{2d}^A(i)w_2(j)\Delta net_{1d}(i, j)f_1'' [net_1(i, j)] + \delta_{2d}^B(i)w_2(j)f_1' [net_1(i, j)]$$

Finally,

$$\begin{aligned} \Delta \bar{\bar{J}}_1 &= \frac{\partial E_d}{\partial \bar{\bar{w}}_1} = -\frac{2}{P} \left\{ \left(\bar{\bar{\delta}}_{1d}^A \right)^T \Delta \bar{\bar{a}}_{1d} + \left(\bar{\bar{\delta}}_{1d}^B \right)^T \bar{\bar{a}}_1 \right\} \\ &\quad \text{with } \bar{\bar{\delta}}_{1d}^A = \bar{\delta}_{2d}^A \bar{w}_2 \cdot * f_1' \left[\overline{\overline{net}}_1 \right] \\ \bar{\bar{\delta}}_{1d}^B &= \bar{\delta}_{2d}^A \bar{w}_2 \cdot * \Delta \overline{\overline{net}}_{1d} \cdot * f_1'' \left[\overline{\overline{net}}_1 \right] + \bar{\delta}_{2d}^B \bar{w}_2 \cdot * f_1' \left[\overline{\overline{net}}_1 \right] \end{aligned} \tag{A.4}$$

APPENDIX B

PULSED RF CALIBRATION FOR NVNA MEASUREMENTS

Pulsed RF large signal measurements are greatly beneficial for the characterization of devices affected by low frequency dispersions and the design of power amplifiers for communication signals with large peak to average power ratio. Pulsed-IV and pulsed-RF measurement techniques enable one to measure a device under iso-thermal and iso-trapping conditions when the pulse duration and duty-rate are selected to be small enough for the low frequency memory effects of the device not to respond.

Recently a promising multiple recording technique was introduced for the LSNA achieving zero pulse desensitization for arbitrary duty-rates. However, the LSNA is typically calibrated using a CW harmonic phase reference (HPR) and CW power meter which do not account for the potential low-frequency dispersions of the LSNA under pulsed operation. In sub-sampling down converter NVNAs such as the LSNA these low-frequency dispersions are associated with the anti-aliasing low pass filter which is used after the samplers.

B.1 Calibration Procedure

In this chapter a calibration procedure is proposed for pulsed-RF measurements with an NVNA. The approach is based on the multi-harmonic characterization of the LSNA, RF pulser and HPR under pulsed operation. The pulsed-RF calibration technique is facilitated

by the fast switching and settling of the pulser which is assumed to have reached quasi-steady state when the RF signals are sampled. First the LSNA is calibrated in CW mode. Then various CW and pulsed-RF measurements are performed to calibrate the LSNA, RF-pulser and pulsed-HPR for pulse operation.

B.1.1 Through Calibration

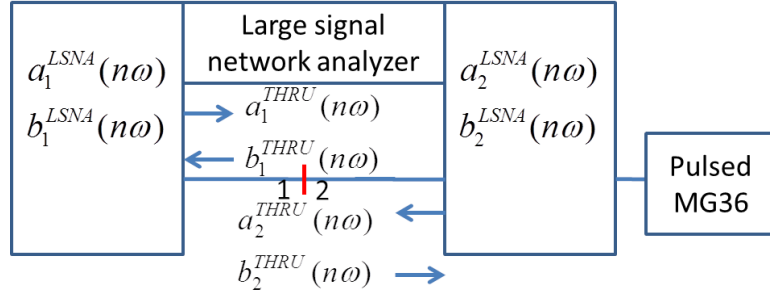


Figure B.1: Measurement setup for a "Through".

The first measurement conducted is that of the through connection shown in Fig B.1. The measured LSNA incident and reflected waves and the actual incident and reflected waves at port 1 and 2, respectively, are related by the following C^{LSNA} dispersion factors :

$$a_{2,P}^{LSNA}(n\omega) = C_{2a,P}^{LSNA}(n\omega)a_{2,P}^{THRU}(n\omega) \quad (B.1)$$

$$b_{1,P}^{LSNA}(n\omega) = C_{1b,P}^{LSNA}(n\omega)b_{1,P}^{THRU}(n\omega) \quad (B.2)$$

The resulting transmission coefficients for the through is then :

$$\frac{b_{1,P}^{LSNA}(n\omega)}{a_{2,P}^{LSNA}(n\omega)} = \frac{C_{1b,P}^{LSNA}(n\omega)b_{1,P}^{THRU}(n\omega)}{C_{2a,P}^{LSNA}(n\omega)a_{2,P}^{THRU}(n\omega)} \quad (B.3)$$

$$= \frac{C_{1b,P}^{LSNA}(n\omega)}{C_{2a,P}^{LSNA}(n\omega)} = T_{12,P-P}^{THRU}(n\omega) \quad (B.4)$$

Since we have $\frac{b_{1,P}^{THRU}(n\omega)}{a_{2,P}^{THRU}(n\omega)} = 1$ for a through measurement, this measurement gives us

$$T_{12,P-P}^{THRU}(n\omega) = \frac{C_{1b,P}^{LSNA}(n\omega)}{C_{2a,P}^{LSNA}(n\omega)}.$$

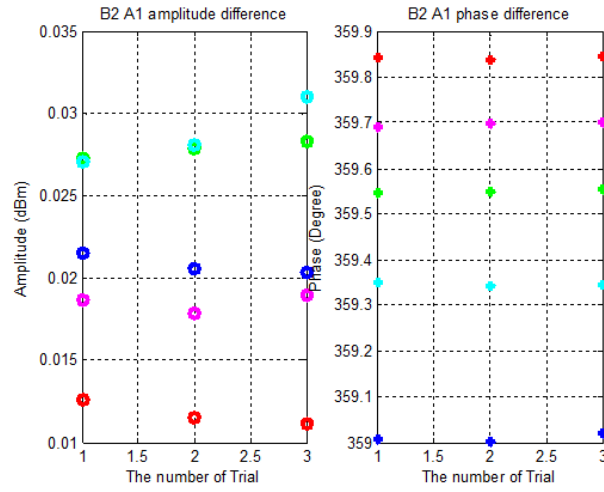


Figure B.2: Amplitude and phases for the through measurements for CW operation for 3 successive.

Fig. B.2 and B.3 shows the amplitude and phase difference with CW and pulsed RF excitations for the "through". Note that we use from top to bottom red for 2 GHz, magenta for 4 GHz, green for 6 GHz, cyan for 8 GHz, and blue for 10 GHz. The LSNA amplitude and phase error corrections for the "through" under CW operation are small: less than 0.01 dB error in amplitude from 2 to 10 GHz and less than 1° error in phase up to 10 GHz. On the other hand the LSNA amplitude and phase error corrections for the "through" under

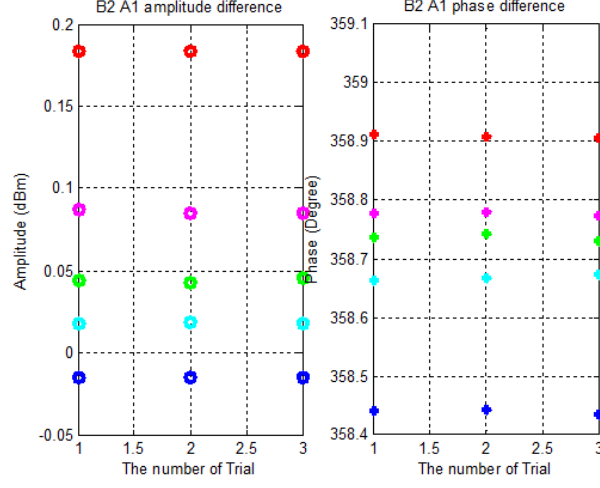


Figure B.3: Amplitude and phases for the through measurements for pulsed operation for 3 successive.

pulsed-RF operation are found to be substantially larger: up to 0.18 dB error in amplitude at 2 GHz and 1.6° error in phase for 10 GHz.

B.1.2 CW-HPR Charcterization

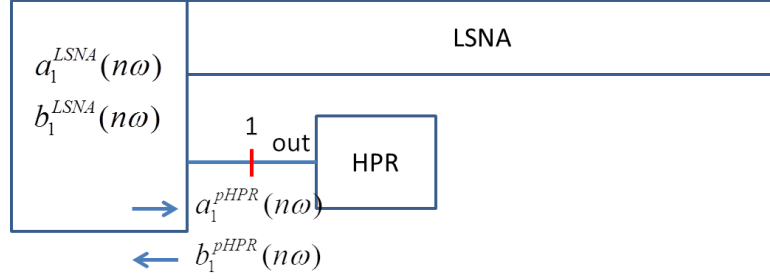


Figure B.4: Measurement setup for the CW-HPR characterization.

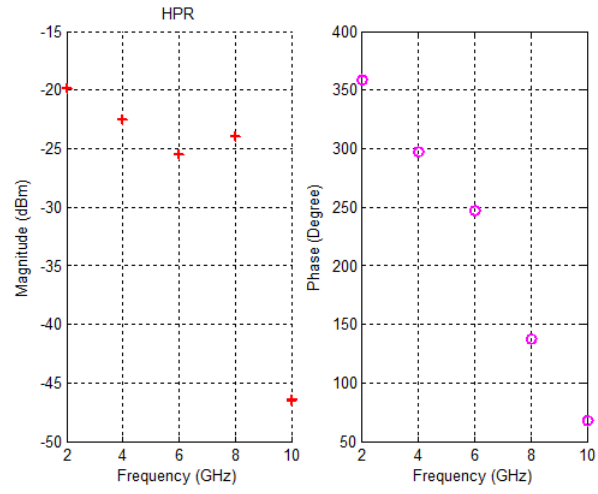


Figure B.5: HPR results under CW excitation up to 5 harmonics for 3 successive measurements HPR amplitude and phase.

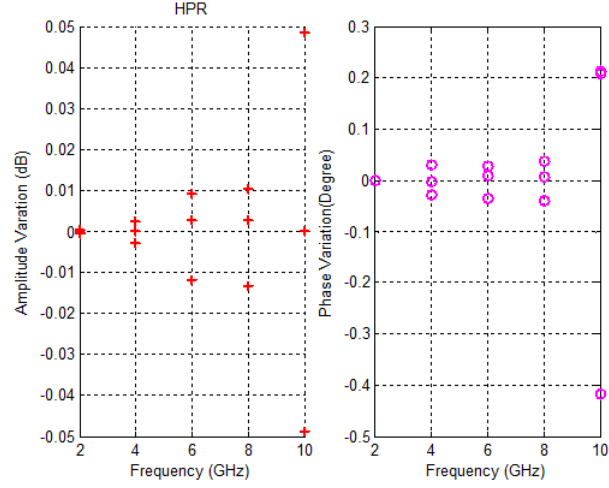


Figure B.6: HPR results under CW excitation up to 5 harmonics for 3 successive measurements HPR amplitude and phase variations.

Next the CW characterization of the HPR is performed. The HPR used is a user-made class-C amplifier with an active biasing circuit. Fig. B.4 shows the testbed used. Fig. B.5 gives the amplitude and phase of the first five harmonics generated. Fig. B.6 compares the fluctuation of the amplitude and phase relative to the fundamental frequency for the first five harmonics. The relative CW phase of the various HPR harmonics are also tabulated in the 3rd column of Table B.1.

The HPR used is seen to provide reasonably stable amplitude and phase for the first five harmonics. The largest error occurs for the 5th harmonic with 0.05 dB amplitude fluctuation and 0.2° phase fluctuation. This is related to its reduced power which is 27 dBc below the fundamental tone.

B.1.3 Pulser Characterization

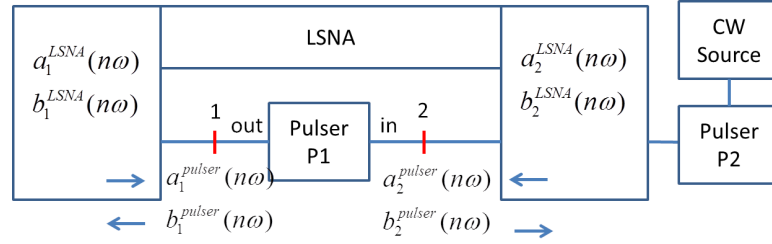


Figure B.7: Measurement setup for the RF pulser characterization.

The measurement setup for the pulser characterization under multiple recording measurement is shown in Fig. B.7. The measured LSNA incident and transmitted waves and the actual incident and transmitted CW and pulsed HPR waves at port 1 and 2 respectively are related by :

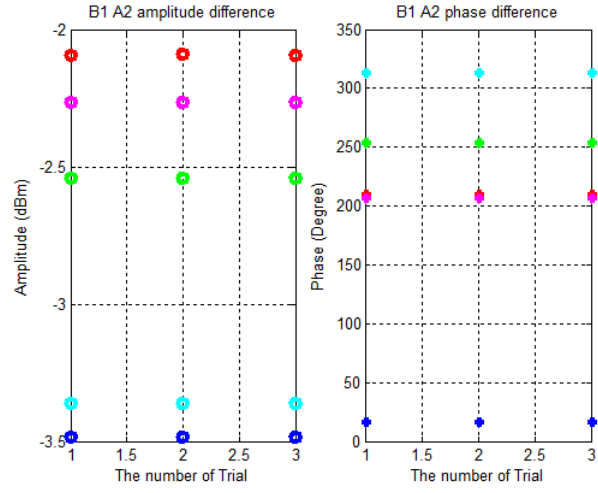


Figure B.8: Amplitude and phase of the S-parameters obtained for the RF pulser at 5 different harmonics for 3 different successive measurements for S_{21} .

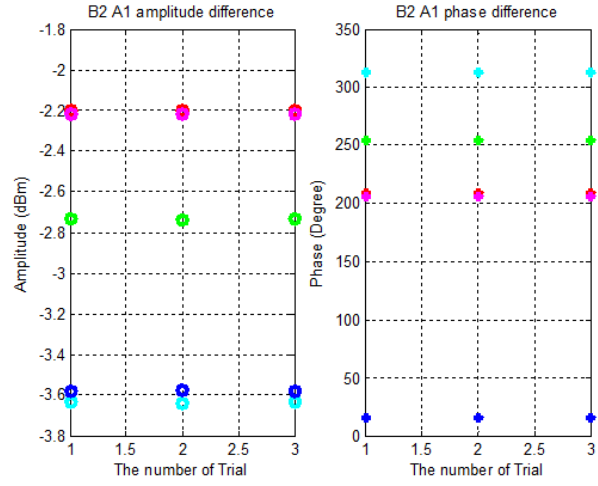


Figure B.9: Amplitude and phase of the S-parameters obtained for the RF pulser at 5 different harmonics for 3 different successive measurements for S_{12} .

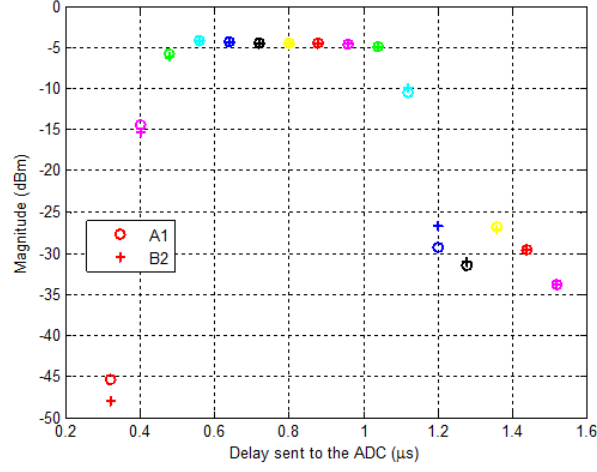


Figure B.10: Alignment of the RF pulses at port 1 and 2.

$$a_{2,P2}^{LSNA}(n\omega) = C_{2a}^{LSNA}(n\omega)a_{2,P2}^{Pulser}(n\omega) \quad (B.5)$$

$$b_{1,P1}^{LSNA}(n\omega) = C_{1b}^{LSNA}(n\omega)b_{1,P1}^{Pulser}(n\omega) \quad (B.6)$$

Fig. B.8B.9 gives the HPR results under pulsed-RF excitation up to the 5th harmonic among 3 successive measurements. Fig. B.8 is for the HPR magnitude and phase and Fig. B.9 is for HPR magnitude and phase variation among 3 successive measurements. The ratio is :

$$\frac{b_{1,P1}^{LSNA}(n\omega)}{a_{2,P2}^{LSNA}(n\omega)} = \frac{C_{1b,P1}^{LSNA}(n\omega)b_{1,P1}^{THRU}(n\omega)}{C_{2a,P2}^{LSNA}(n\omega)a_{2,P2}^{THRU}(n\omega)} \quad (B.7)$$

$$= T_{12,P-P}^{THRU}(n\omega) \frac{b_{1,P1}^{Pulser}(n\omega)}{a_{2,P2}^{Pulser}(n\omega)} = T_{12,P-P}^{THRU}(n\omega) S_{12,P1-P2}^{Pulser} \quad (B.8)$$

using the assumption of the same pulsers being used : $P1 \approx P2$. :

$$T_{12,P1-P2}^{THRU}(n\omega) = \frac{C_{1b,P1}^{LSNA}(n\omega)}{C_{2a,P2}^{LSNA}(n\omega)} \approx \frac{C_{1b,P1}^{LSNA}(n\omega)}{C_{2a,P1}^{LSNA}(n\omega)} = T_{12,P1-P1}^{THRU}(n\omega) \quad (B.9)$$

It results that the pulser parameter $S_{12,P1-P2}^{Pulser}(n\omega) = \frac{b_{1,P1}^{Pulser}(n\omega)}{a_{2,P2}^{Pulser}(n\omega)}$ is obtained from this measurement.

The derivation above assumes that the system is reasonably well matched. An extension of the calibration theory for the case of miss-matched pulsers and sources will reported elsewhere.

The results obtained for the pulser S-parameters under pulsed-RF excitations are shown in Fig. B.8B.9. The forward S_{21} and backward S_{12} are seen to be similar indicating that the pulser is symmetric. The relative phase of S_{21} for the linear filter are also tabulated in the 2nd column of Table B.1. The associated relative phase-shift for the harmonics are defined relative to the fundamental tone using :

$$\phi^{relative}(n\omega) = \phi(n\omega) - n\phi(\omega) \quad (\text{B.10})$$

Note that the alignment between the incident and reflected waves measured by the LSNA is performed using the usual procedure as shown in Fig. B.10 by changing the delay of the multiple recording trigger sent to the ADC.

B.1.4 Pulser Characterization

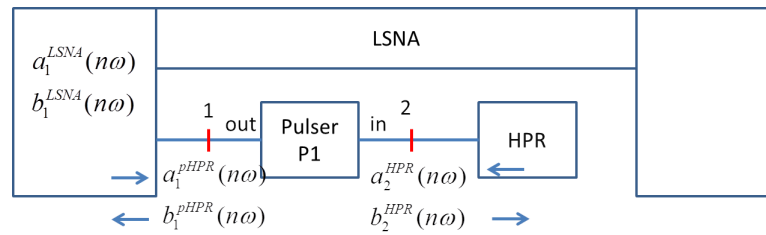


Figure B.11: Measurement setup for the characterization of the pulsed HPR.

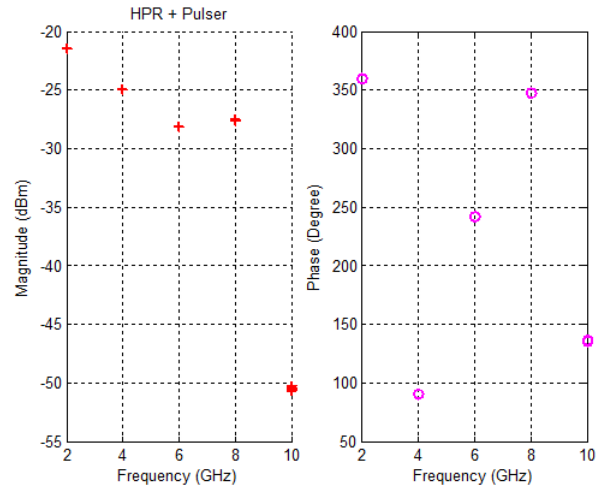


Figure B.12: Pulsed-HPR results up to 5 harmonics among 3 successive measurements : : pulsed-HPR amplitude and phase.

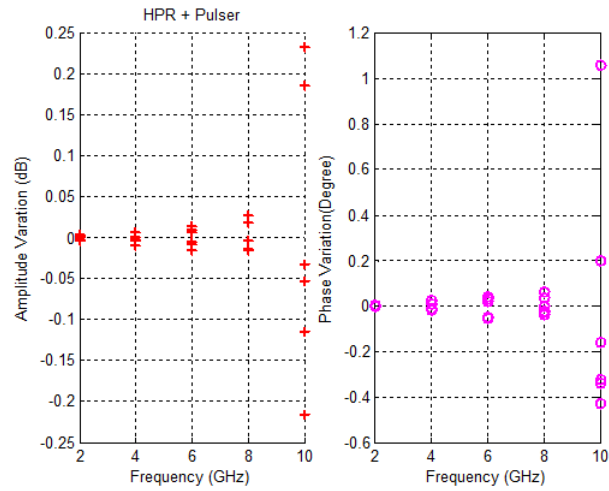


Figure B.13: Pulsed-HPR results up to 5 harmonics among 3 successive measurements : : pulsed-HPR amplitude and phase variation.

The measurement setup for the pulsed-HPR characterization under multiple recording measurements is shown in Fig. B.11. The measurement LSNA incident wave and the actual incident waves at port 1 are related by :

$$b_{1,P1}^{LSNA}(n\omega) = C_{1b,P1}^{LSNA}(n\omega)b_{1,P1}^{pHPR}(n\omega) \quad (B.11)$$

$$b_{1,P1}^{LSNA}(n\omega) = S_{12,P1-CW}^{Pulser}(n\omega)a_{2,CW}^{HPR}(n\omega) \quad (B.12)$$

This measurement give us (after the relative phase correction) :

$$\frac{b_{1,P1}^{LSNA}(n\omega)}{b_{1,P1}^{LSNA}(\omega)} = \frac{C_{1b,P1}^{LSNA}(n\omega)b_{1,P1}^{pHPR}(n\omega)}{C_{1b,P1}^{LSNA}(\omega)b_{1,P1}^{pHPR}(\omega)} \quad (B.13)$$

$$= \frac{C_{1b,P1}^{LSNA}(n\omega)S_{12,P1-CW}^{pulser}(n\omega)a_{2,CW}^{HPR}(n\omega)}{C_{1b,P1}^{LSNA}(\omega)S_{12,P1-CW}^{pulser}(\omega)a_{1,CW}^{HPR}(\omega)} \quad (B.14)$$

Given $S_{12,P1-CW}^{pulser}(n\omega)$ and $\frac{a_{2,CW}^{HPR}(n\omega)}{a_{1,CW}^{HPR}(\omega)}$ are known we obtain the HPR calibration correction factor : $\frac{C_{1b,P1}^{LSNA}(n\omega)}{C_{1b,P1}^{LSNA}(\omega)}$. We assume : $S_{12,P1-CW}^{pulser}(n\omega) \approx S_{12,P1-P2}^{pulser}(n\omega)$. This is equivalent to assuming the broadband pulser used has no low-frequency dispersion during the measurement interval. Fig. B.6 compares the fluctuation of the amplitude and phase relative to the fundamental frequency for the first five harmonics of the pulsed-HPR. The relative phase of the various pulsed-HPR harmonics are also tabulated in the 3rd column of Table B.1. Like for the HPR first characterized under CW operation, the HPR under pulsed-RF operation is seen to provide reasonably stable amplitude and phase for the first five harmonics.

Table B.1 shows the measured and expected phase for the 5 harmonics of the HPR and pulser. The sum of the relative phase of the pulser and the relative phase of the CW-HPR calibration gives the expected pulser+HPR relative phase in the 4th column. The difference between the expected and actually measured phases for the Pulser+HPR yields in the last column the calibration phase correction due to the LSNA dispersion under pulse operation. It is to be noted that the error appears to be in a linear progression with the harmonic number

Table B.1: Measured and Expected Phase for the 5 Harmonics of the HPR and Pulser

Harmonics	Pulser(°)	HPR(°)	Expected Pulser + HPR (°)	Measured Pulser + HPR (°)	Correction factor (°)
1 st	0	0	0	0	0
2 nd	148.2± 0.0	299.4± 0.0	87.6± 0.0	90.3± 0.0	-2.7
3 rd	346.3± 0.0	249.8± 0.0	236.1± 0.0	241.7± 0.1	-5.6
4 th	196.7± 0.0	141.9± 0.0	338.6± 0.0	346.7± 0.1	-8.1
5 th	51.3± 0.0	72.8± 0.4	124.1± 0.4	135.3± 1.1	-11.2

which suggests that the LSNA dispersion under pulsed operation is simply associated with the group delay of the anti-aliasing low-pass filter.

B.2 Results for the Power Calibration

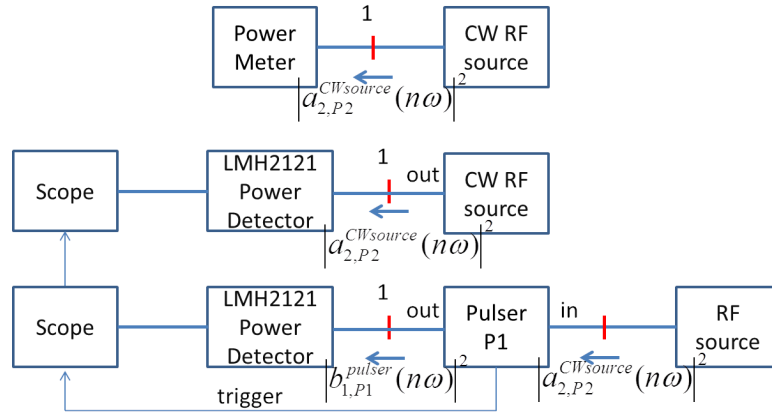


Figure B.14: Measurement setup for power calibration.

The absolute power calibration of the LSNA requires the independent measure of the pulsed RF power using the measured characteristics of the pulser:

$$\left[b_{1,P1}^{pHPR}(n\omega) \right]^2 = [S_{12,P1-P2}^{pulser}]^2 [a_{2,P2}^{CW_{source}}(n\omega)]^2. \quad (\text{B.15})$$

This power calibration can then be completed using the pulsed-HPR. Alternately the pulsed-RF power calibration can be performed using a fast power detector as outlined in Fig. B.14. First a CW RF source is calibrated with a power meter to provide 1/0.1 mW for each of the harmonics. Next a fast RF envelope detector is calibrated using this RF power source operating at the same power level. The envelope detector is then used to measure the pulsed RF power and the CW RF power is tuned till the correct envelope response for 1/0.1 mW is obtained. This simple power calibration assumes that the pulser exhibits fast switching and settling characteristics. This required feature is readily verified by inspecting the flatness of the response of the envelope detector within the RF pulse.

Table B.2 shows the results obtained using this procedure. As shown in Table B.2, the difference between the calibrated (0 dBm or -10 dBm) and the measured values for the power at the various harmonic yields in the last column the calibration power correction due to the LSNA dispersion under pulse operation. A small calibration correction is found to be required due to the dispersion of the buffer stage (low-pass filter) between the LSNA samplers and ADCs.

B.3 Conclusion

In this chapter a new calibration theory for pulsed-RF large signal measurement has been presented. The approach relies on a new pulsed HPR and pulsed power calibration beside the conventional linear calibration. The methodology used is based on the multi-harmonic characterization of the RF pulser under pulsed operation. The technique was

Table B.2: Difference Between Calibrated And Measured Values

Harmonics	Signal Source Setting (dBm)	Power Meter (dBm)	Oscilloscope (mV)	Pulsed LSNA Measurements (dBm)	Correction factor (dB)
1 st	3.97	0.00	14.72	0.60	0.60
2 nd	5.88	0.00	15.12	0.65	0.65
3 rd	8.13	0.00	15.57	1.00	1.00
4 th	11.64	0.00	15.07	1.40	1.40
5 th	9.95	-10.00	2.47	-7.60	2.40

demonstrated for an LSNA operating using the multiple recording technique. The proposed pulsed-RF calibration approach could also be applied to other NVNAs.

APPENDIX C

LARGE SIGNAL ANALYSIS

C.1 Large-signal analysis

C.1.1 Small-signal analysis

For the S-parameters measurements of a device, two separate measurements which provide the ratios of the incident and reflected waves under the excitation of a single tone, are performed with the vector network analyzer (VNA). The single tone is applied at the input port while the output port is terminated by 50 Ohms, and then the single tone is applied at the output port with 50 Ohms termination at the input port, successively. By solving the resulting system of linear equations, the ratios can be evaluated yielding the S-parameters at the single frequency.

Note that this small-signal analysis relies on the superposition principle. Thus the stimulus has to be sufficiently small in order to keep the device operating in its linear operating region.

C.1.2 Large-signal analysis

Small-signal analysis requires that the power level of the signal be small enough for the device under test (DUT) to respond linearly to the excitation provided at both ports. Many

applications such as power amplifiers require significantly higher signal levels to maximize their efficiency such that the device operates in non-linear regime. For such application, S-parameters are no longer sufficient in non-linear region because the S-parameters are only valid in the linear regime.

Moreover, in the non-linear regime, the superposition principle can be used to analyze the DUT response. Therefore, it is crucial to be able to put the DUT under realistic large-signal operating conditions to acquire complete and accurate information. OSU has acquired a large signal network analyzer (LSNA) in Spring 2004 under an AFOSR equipment grant for performing large-signal network analysis.

C.1.3 LSNA system

The unique advantage of the LSNA is its ability to make vectorial characterization of the non-linear response of RF systems under periodic signals excitation so that both the time and frequency domain response of the device be evaluated. The LSNA has the unique ability of measuring both the absolute phase and the amplitude of interesting frequencies at both the fundamental as well as the harmonics up to 50 GHz with unique calibration schemes, e.g. power and phase calibrations, namely.

When a signal consists of N harmonics, it can be represented by a discrete set of tones in the frequency domain by using the following equation

$$x(t) = \text{Re} \left\{ \sum_{i=0}^N X_i e^{j\omega_i t} \right\}. \quad (\text{C.1})$$

Also, the RF signals can be either modulated or pulsed. Up to more than 10,000 RF tones can be acquired allowing for pulsed-RF measurements or UWB measurements of periodic signals at single measurement within 20 MHz. With M modulated tones in the single-side

band (SSB) and N harmonics, the signal can be expressed as

$$x(t) = \text{Re} \left\{ \sum_{i=0}^N \sum_{k=-M}^M X_{ik} e^{j\omega_k t} e^{j\omega_i t} \right\}. \quad (\text{C.2})$$

An example of the measured signal is represented in Fig. C.1. A two-tone signal with the tone spacing of 10 kHz is applied at f_0 of 2 GHz. In (a) and (b), the white lines and red dots represent the absolute magnitude and the phase of each frequency components, respectively. In (b), intermodulation components are shown by DUT even though the injected signal contains only two dominant tones. The time-domain signal in (c) shows the carrier signal (yellow) along with the individual harmonic envelopes.

Essentially the LSNA measures the incident (a) and reflected (b) waves applied to the DUT at the reference planes which are established at calibration. These can then be converted into the voltage (v) and the current (i) waves. The equations show below establish the relationship between v, i and a, b of both planes.

$$A = \frac{V + Z_c I}{2} \quad (\text{C.3})$$

$$B = \frac{V - Z_c I}{2} \quad (\text{C.4})$$

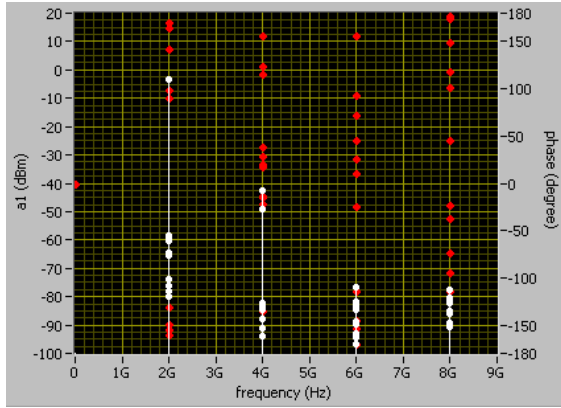
and,

$$V = A + B \quad (\text{C.5})$$

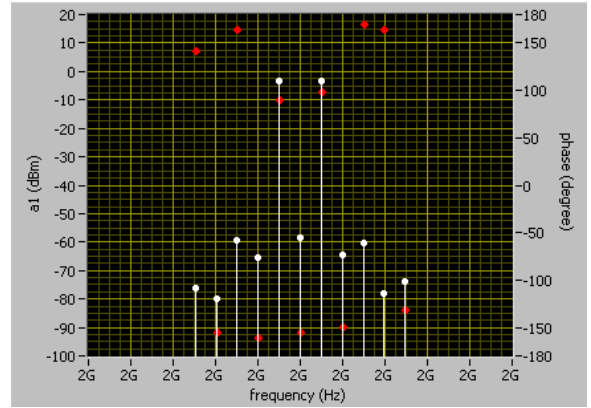
$$I = \frac{A - B}{Z_c} \quad (\text{C.6})$$

where, in general, Z_c is 50Ω .

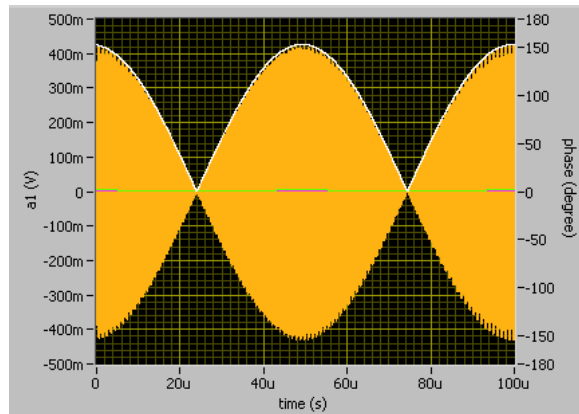
The basic system of LSNA consists of a test set, a sampling converter, a signal generator, a power meter and an harmonic phase reference (HPR) module. The test set separates the incident and the reflected waves at both ports 1 and 2 using internal couplers. The original couplers used have a coupling factor of 14 dB. The resulting four channels are then



(a) Two-tone signal with 4 harmonics



(b) Two-tone signal at f_0



(c) Two-tone signal in time

Figure C.1: Two-tone signal measurements with the LSNA; White solid line is magnitude and red dot is phase at each frequency components

attenuated by step attenuators, which can vary from 10 dB to 40 dB, in order to protect the sampling converter from excessive power as well as keep the LSNA operating in its linear regime. Then, the attenuated waves are connected to the sampling converter for each channel, which converts the RF signals into IF signals using the sub-harmonic downversion sampling method using an LO frequency established by the sampling SDR diode. The down-converted IF signals are then acquired by the analog-digital-converter (ADC) which yield voltage (v) and the current (i) waves of the DUT.

In addition to the standard calibration method (SOLT or LRRM) typically used in normal network analyzers, the LSNA calibration needs additional information to determine the absolute power and phase in the frequency range of interest. The power meter (sensor) and HPR provides those extra information. The HPR is a step recovery diode (SRD) which generates a very stable pulse that contain a lot of harmonic frequencies of the LO frequency. Therefore, it provides phase information for harmonic frequencies of interest relative to the fundamental frequency, which is also generated by the SRD.

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